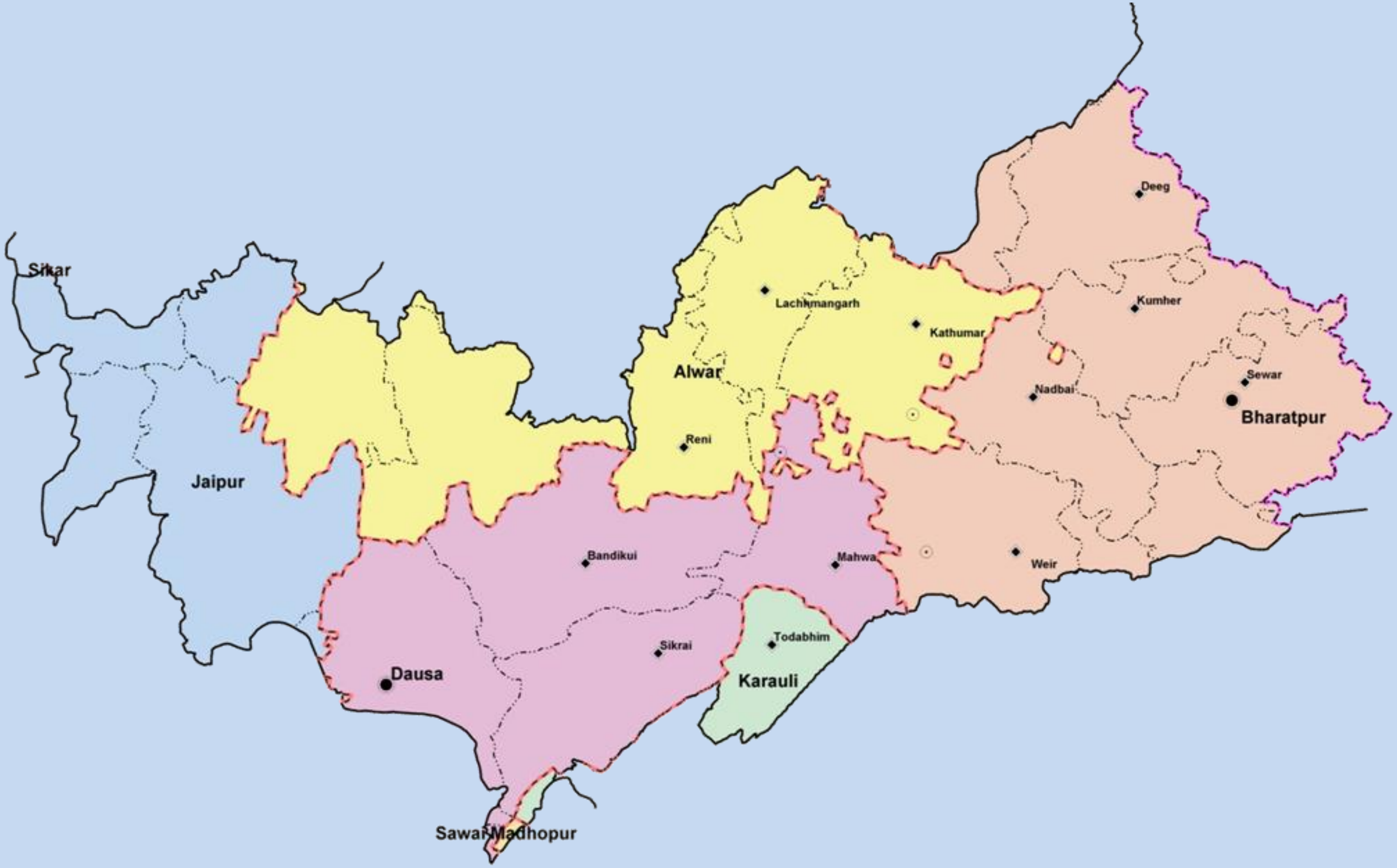
Ground Water Department, Rajasthan

European Union
State Partnership Programme

Hydrogeological Atlas of Rajasthan

Banganga River Basin



2013

Rolta India Limited

Message

सत्यमेव जयते

Shri Ashok Gehlot,
Hon'ble Chief Minister, Rajasthan

Ground water is an inseparable aspect in the lives of the people of Rajasthan. There are no major or perennial rivers flowing through the State. Moreover, most part of the state receives very less rainfall. Our dependence on ground water is extremely critical not only as drinking water for human beings and cattle but also as major source of irrigation.

The Government of Rajasthan is keen on sustainable utilization of the ground water. The State water policy, adopted in 2010 emphasizes on community participation in ground water management and use of information technology for widespread dissemination of data for the benefit of all sections of the society.

I am pleased to note that the Ground Water Department of the State has carried a detailed assessment of quality, quantity and distribution of ground water resources of the State through mapping of aquifers and using most modern tools and techniques.

It is also heartening to note that the outcome of the project is being published in the form of 'Hydrogeological Atlas of Rajasthan'. It will incorporate the scientific findings and reliable information in various types of readily understandable GIS maps based on ground water potential viz. zone maps, depth to water level maps, ground water categorization maps, chemical quality maps, aquifer maps etc. These would be certainly found useful by experts and user agencies. The same would also help in raising the awareness levels of the various stakeholders in utilizing the precious ground water resources of the State.

I take this opportunity to congratulate the Ground Water Department officials for bringing this useful publication and wish them success in all such endeavor.

(Ashok Gehlot)

Message

सत्यमेव जयते

Dr. Jitendra Singh,
Hon'ble Minister for Energy,
Non-Conventional Energy Resources,
PHED, GWD, Information & Public Relations

Ground water is a renewable resource but its long term sustainability dependents on the balance of recharge and withdrawal ground water is added every year through rainfall and other sources. In Rajasthan, volume of water extracted for different purposes far exceeds its replenishment which has led to depletion of ground water in large parts of state.

While the surface water is visible and largely measurable, the ground water is hidden underneath. To meet the ever increasing demand of water, the state has been promoting conjunctive use of water. In order to have a state level estimation of water resources, the estimation of ground water quality and quantity is a key input. The studies being very scientific in nature it remains a challenge how to communicate the message to all stakeholders.

The publication of '*Hydrogeological Atlas of Rajasthan'* is a major step forward and will help in visualization of ground water related information in very simplified graphic and tabular forms. The effort that has gone into creation of the atlas from basic information in archives is very much appreciated.

15.4.13.

(Jitendra Singh)

Message

सत्यमेव जयते

Dr. Purushottam Agarwal
Principal Secretary to Government,
Public Health Engineering and Ground Water Department, Rajasthan

Ground water has been the principal source over years to meet the needs of human and cattle population of the state. Increasingly modern techniques are used for extraction of large volume of ground water which has exceeded recharge in most parts of the State. As a result, water table in most of the areas has gone down significantly. People are digging deeper and deeper to reach the ground water to meet their increasing needs. Maintaining a healthy practice of balance between recharge and withdrawal is critical for its sustainability.

The state had initiated various programs under which a systematic assessment of water resources is being carried out and all future water management plans are chalked out on the basis of sound database and historic data analysis. Increasing population, industrialization, agricultural need, power generation etc. are bound to pose challenges before us in coming years and to face them we have documented 'Water Resource Vision 2045' which highlights the short term (upto 2015) and long term (upto 2045) thrust areas and action plan which are pre-requisites for successful implementation of the State Water Policy. We are committed to achieving the objective of optimum use of every drop of scarce and precious utilizable water resource. In this regard, the Ground Water Department has also carried out a systematic assessment of ground water resources and three dimensional mapping of aquifers that gives a new insight.

I congratulate the Ground Water Department on publication of 'Hydrogeological Atlas of Rajasthan' which emphasizes on various aspects of ground water which I feel will help us in spreading the information on current scenario and consequently help in managing it better.

(Purushottam Agarwal)

Foreword

European Union
State Partnership Programme

Arno Schaefer
Minister Counselor, Head of Operations

The Government of Rajasthan has entered into an agreement with the European Union through the State Partnership Programme for implementing state-wide water sector reform leading to sustainable integrated water resources management in Rajasthan, and supporting the Panchayat Raj institutions in executing their responsibilities in equitable access to safe, adequate, affordable and sustainable water supply as well as conservation, stabilization and replenishment of surface and ground water.

Ground water is the primary source for drinking, domestic, agriculture and industries and other purposes in Rajasthan. The stage of ground water development is close to 135% i.e., if the net annual ground availability in Rajasthan is about 10.7 billion cubic meter, the annual draft is 14.5 billion cubic meter. As a result of this, the water levels have been depleting alarmingly. The ground water in 166 out of 249 blocks of the state fall under the category of over-exploited. This indiscriminate and sometime exploitative use of ground water has led to questions regarding its sustainability.

This publication '*Hydrogeological Atlas of Rajasthan*' is the culmination of a long term effort of collation, integration, interpretation and analysis of vast archive of historic data obtained by Rajasthan Ground Water Department. The atlas is in three parts, (1) Statewide Atlas, which shows the state of ground water, water quality, current stage of exploitation, aquifer mapping and basic information on administrative setup, climate, geology and geomorphology; (2) A detailed series, wherein compilation of information and district level thematic maps and all 33 districts have been separately presented and (3) Compilation of information at the River Basin level. In addition to the thematic maps, a large number of cross sections and the 3 dimensional features of aquifers have also been presented.

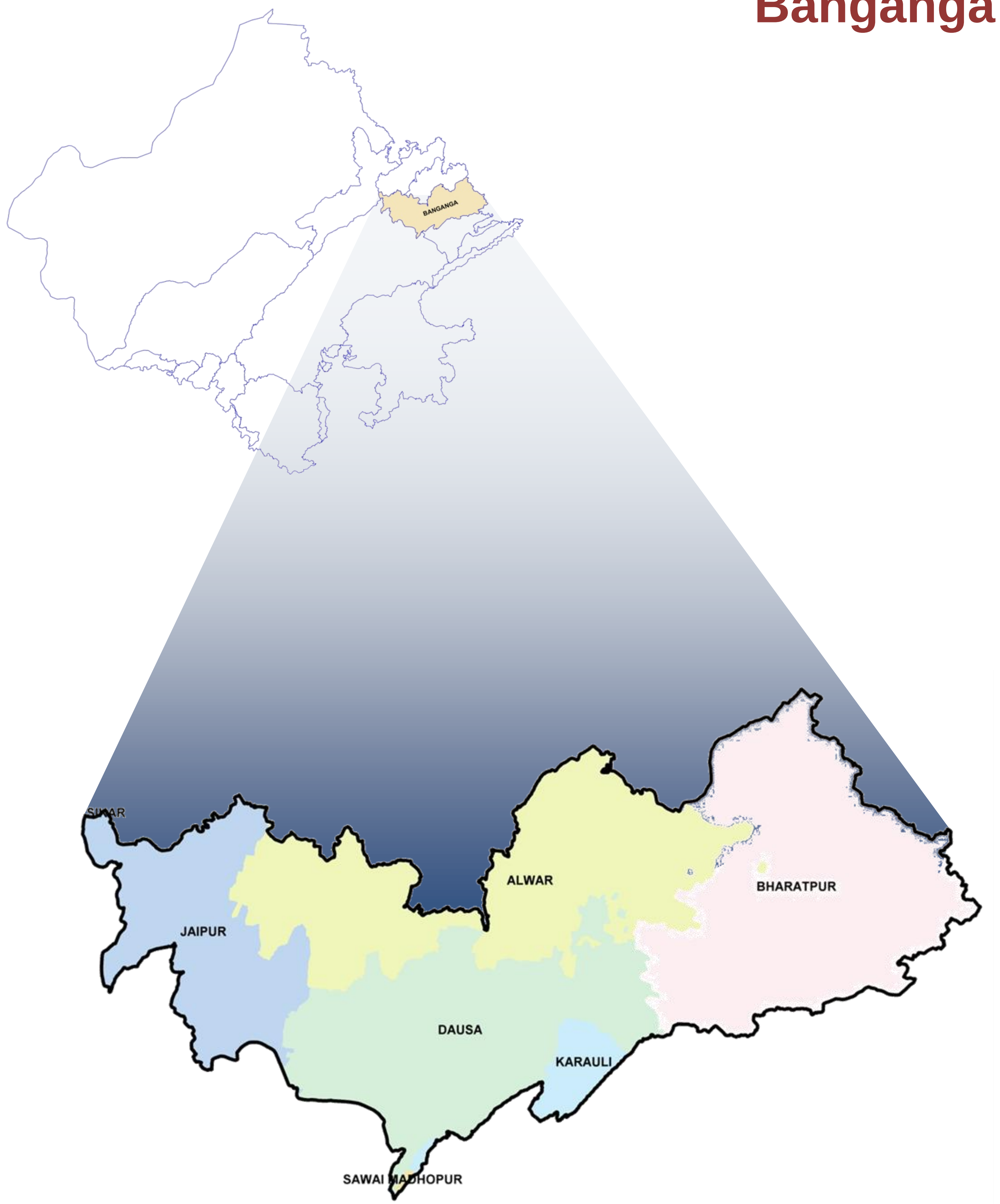
I am convinced that the Atlas will contribute to better understand ground water and help managerial and future planning. The current Geographic Information System (GIS) based study is a pioneering step in India and will pave the way for similar approach to be adopted by other Indian States.

I appreciate very much the efforts made by the Ground Water Department, Government of Rajasthan, The European Union Technical Assistance Team and M/s Rolta India Limited for undertaking this study and compiling this Atlas under the India-EU State partnership Programme.

(Arno Schaefer)

Hydrogeological Atlas of Rajasthan

Banganga River Basin



Contents:

2013

Location:

Banganga River Basin is located in the northeastern part of Rajasthan state, encompassing area between 26° 38' 57.21” to 27° 37' 25.10” North latitudes and 75° 48' 23.37’’ to 77° 41' 31.27’’ East Longitudes. The total catchment area of the Basin is 8,878.7 sq km. It is bounded by the Gambhir and Banas River Basins in its south-southwest; Ruparail and Sabi in its north; and the Shekhawati Basin in its west. Its eastern border is marked the Yamuna River Basin and Uttar Pradesh State.

River Banganga originates in the Aravali hills, near Arnasar and Bairath in Jaipur District. It flows towards the south up to the village of Ghat, then towards east through partly hilly and partly plain terrain. The total length of the river is approximately 240 km. The main tributaries are Gumti Nalla and Suri River, joining the river on its right bank, and Sanwan and Palasan Rivers meeting the river on its left bank. Alluvial deposits cover major part of the basin, while various hardrock formations outcrop in the upstream part of the basin, on both sides of the river, and as scattered and isolated hillocks in alluvial terrain.

Administrative Set-up:

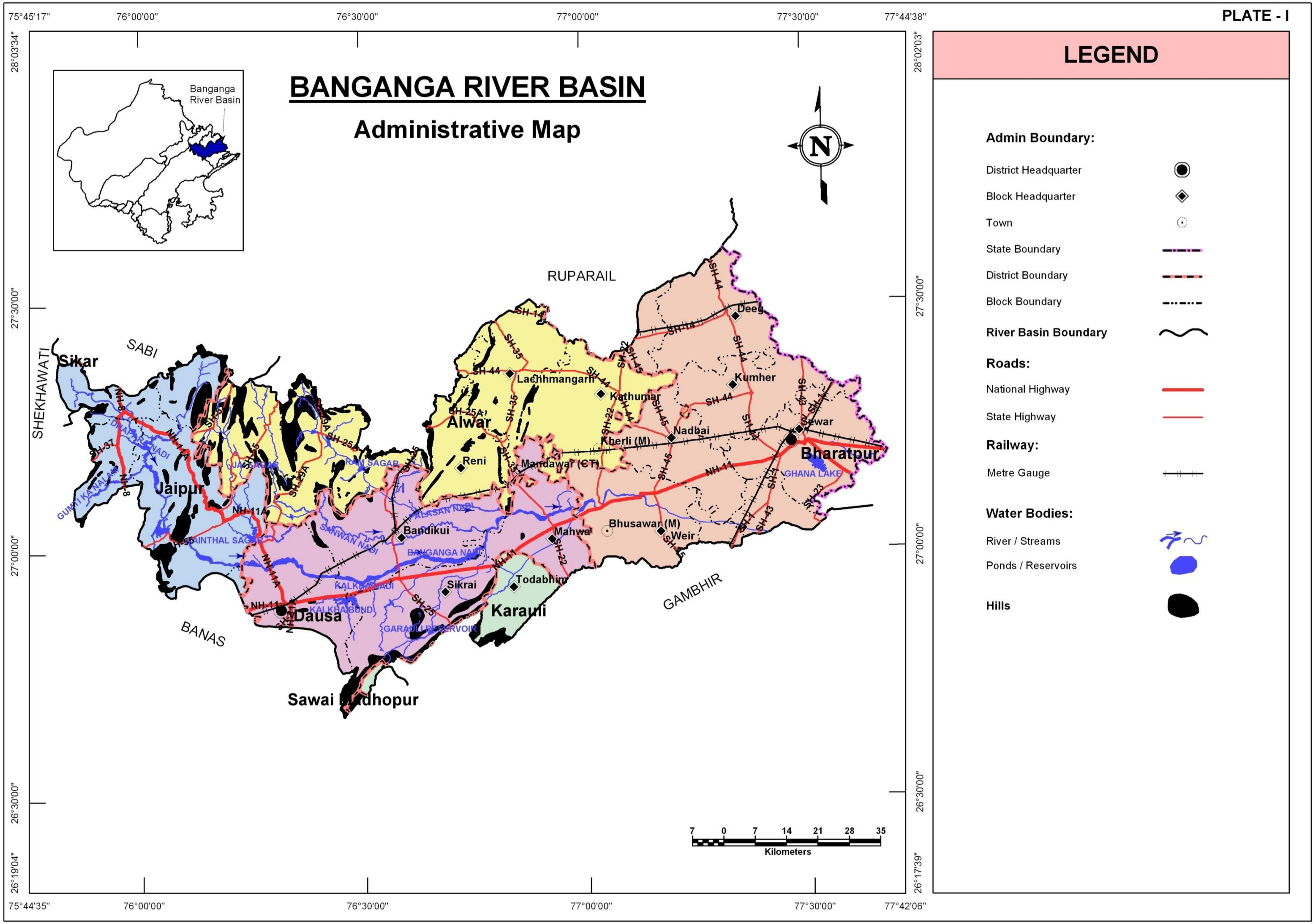
Administrative, Banganga River Basin extends over parts of Alwar, Jaipur, Dausa, Sawai Madhopur and Bharatpur Districts. The total number of fully or partially covered administrative blocks in this basin is approximately 30 and within 2,473 towns and villages.

S. No.	District Name	Area (sq km)	% of Basin Area	Total Number of Blocks	Total Number of Towns and Villages
1	Alwar	2,332.6	26.5	7	594
2	Bharatpur	2,746.8	31.3	9	844
3	Dausa	2,131.8	24.3	5	621
4	Jaipur	1,325.4	15.1	5	101
5	Karauli	244.1	2.8	2	308
6	Sawai Madhopur	7.6	0.1	1	3
7	Sikar	0.4	0.1	1	2
TOTAL		**8,788.7**	**100.0**	**30**	**2,473**

Climate:

Banganga basin’s climate is sub-humid and receives fairly good rainfall during monsoon season. It is very cold from November to February while turns very hot from March to June. The mean annual rainfall over Banganga Basin was 596 mm, of which about 95% is received during the Monsoon months spreading from end of June to September. The transition months from September to November are very pleasant.

PLATE - I


BANGANGA RIVER BASIN
Administrative Map
Banganga River Basin
LEGEND
Admin Boundary:
District Headquarter
Block Headquarter
Town
State Boundary
District Boundary
Block Boundary
River Basin Boundary
Roads:
National Highway
State Highway
Railway:
Metre Gauge
Water Bodies:
River / Streams
Ponds / Reservoirs
Hills
RUPARAIL
SABI
SHEKHAWATI
BANAS
GAMBHIR
Sikar
Jaipur
Alwar
Dausa
Karauli
Sawai Madhopur
Bharatpur
Deeg
Kumher
Nadbai
Kathumar
Lachhmangarh
Reni
Mandawar (CT)
Kherli (M)
Bandikui
Mahwa
Bhusawar (M)
Weir
Sikrai
Todabhim
GHANA LAKE
BANGANGA NADI
SANWAN NADI
KALKHA NADI
GUMTI KA NALA
KALKHA BOND
GARANLI RESERVOIR
NH-11
NH-11A
NH-8
SH-44
SH-35
SH-14
SH-25A
SH-22
Kilometers
7 0 7 14 21 28 35
75°45'17" 76°00'00" 76°30'00" 77°00'00" 77°30'00" 77°44'38"
75°44'35" 76°00'00" 76°30'00" 77°00'00" 77°30'00" 77°42'06"
28°03'34" 27°30'00" 27°00'00" 26°30'00" 26°19'04"
28°02'03" 27°30'00" 27°00'00" 26°30'00" 26°17'39"

TOPOGRAPHY

The western part of the Basin is marked by hilly terrain belonging to the Aravali range, with fairly flat valleys along the Banganga River and its tributaries. East of the Todabhim - Mandawar chain of hills, lies an extensive alluvial plain with gentle easterly slopes towards the Yamuna River in Uttar Pradesh. The northeastern part of the area is also rather flat, interspersed with moderately elevated hills.

Physiographically, the district can be divided into two distinct units, viz. Alluvial plain and Hilly areas. The Western part of the basin where the Banganga originates is largely hilly and further westwards and most of the eastern part is alluvial plain with gently undulating topography. The hills attain a maximum elevation of 727m in Jaipur district and topographically the minimum elevation of 160m is seen in Bharatpur district. The Banganga River is an east flowing river and after flowing through the basin it drains into Ghana Lake near Bharatpur city and thus forming an inland drainage system.

Table: District wise minimum and maximum elevation

District Name	Min. Elevation (m amsl)	Max. Elevation (m amsl)
Alwar	200	689
Bharatpur	160	416
Dausa	222	594
Karauli	233	438
Jaipur	322	727
Sawai Madhopur	430	526
Sikar	575	698

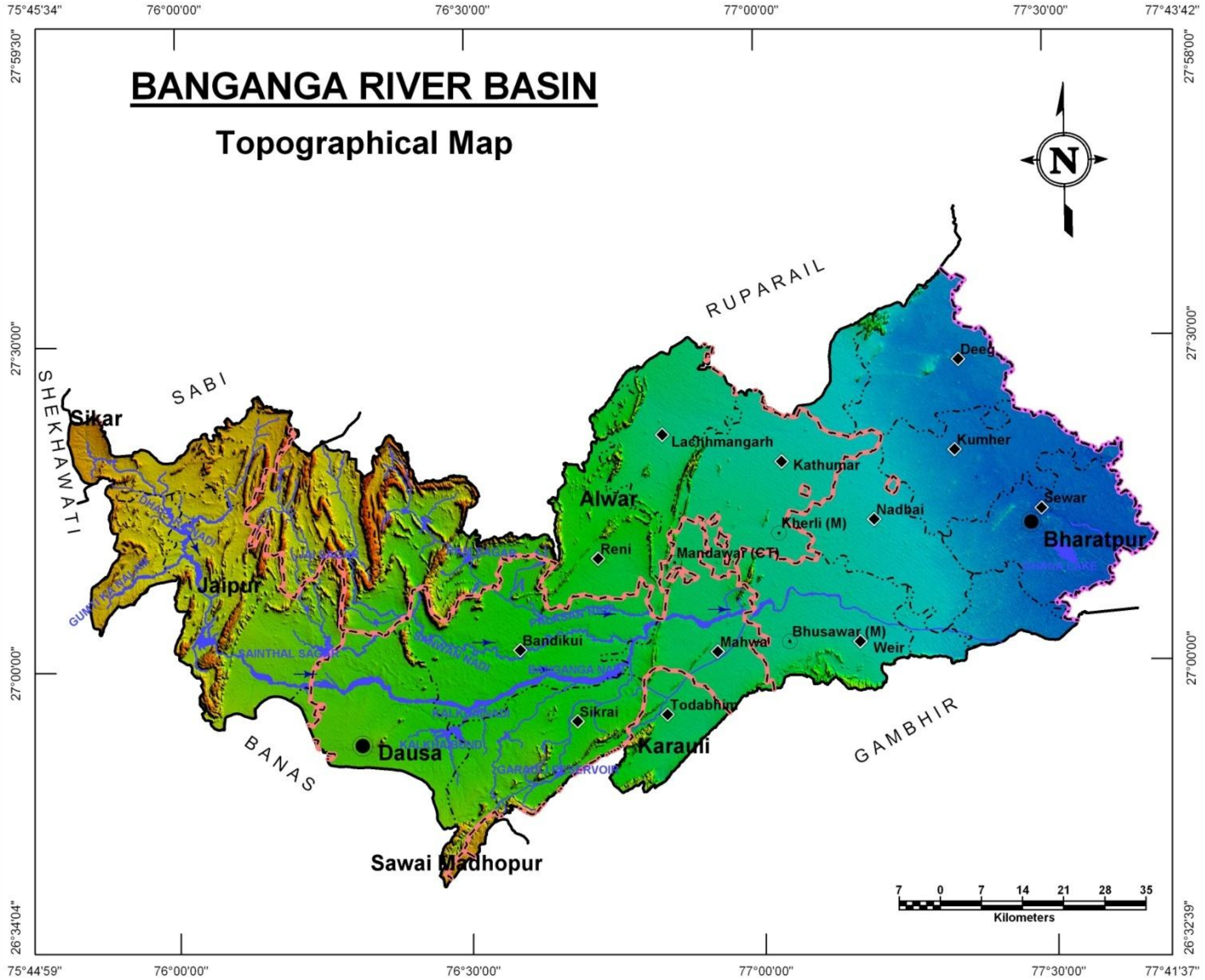
RAINFALL

The rainfall in this basin is fairly good as compared to other parts of Rajasthan. The general distribution of rainfall across can be visualized from isohyets presented in the Plate – III where most of the central part of the basin received rainfall in the range of 500-800mm in year 2010, however reaching a maximum of 1039mm in Jamwa Ramgarh Block in Western hilly part of the basin.

Table: District wise total annual rainfall (based on year 2010 meteorological station recordings (http://waterresources.rajasthan.gov.in)

Rain Gauge Station Location	Total Monsoon Rainfall (mm)	Total Non-Monsoon Rainfall (mm)	Total Annual Rainfall (mm)
Baswa	547	103	650
Bharatpur	684	70	754
Dausa	640	78	718
Deeg	869	58	927
Jamwa Ramgarh	950	89	1,039
Kathumbar	786	48	816
Laxmangarh	679	51	730
Kumbher	722	48	770
Mahuwa	424	117	541
Nadbai	708	53	761
Sikrai	550	110	660
Todabhim	555	73	628
Weir	595	114	709
Bahadurgarh	798	18	816

PLATE - II



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

River Basin Boundary

Water Bodies:

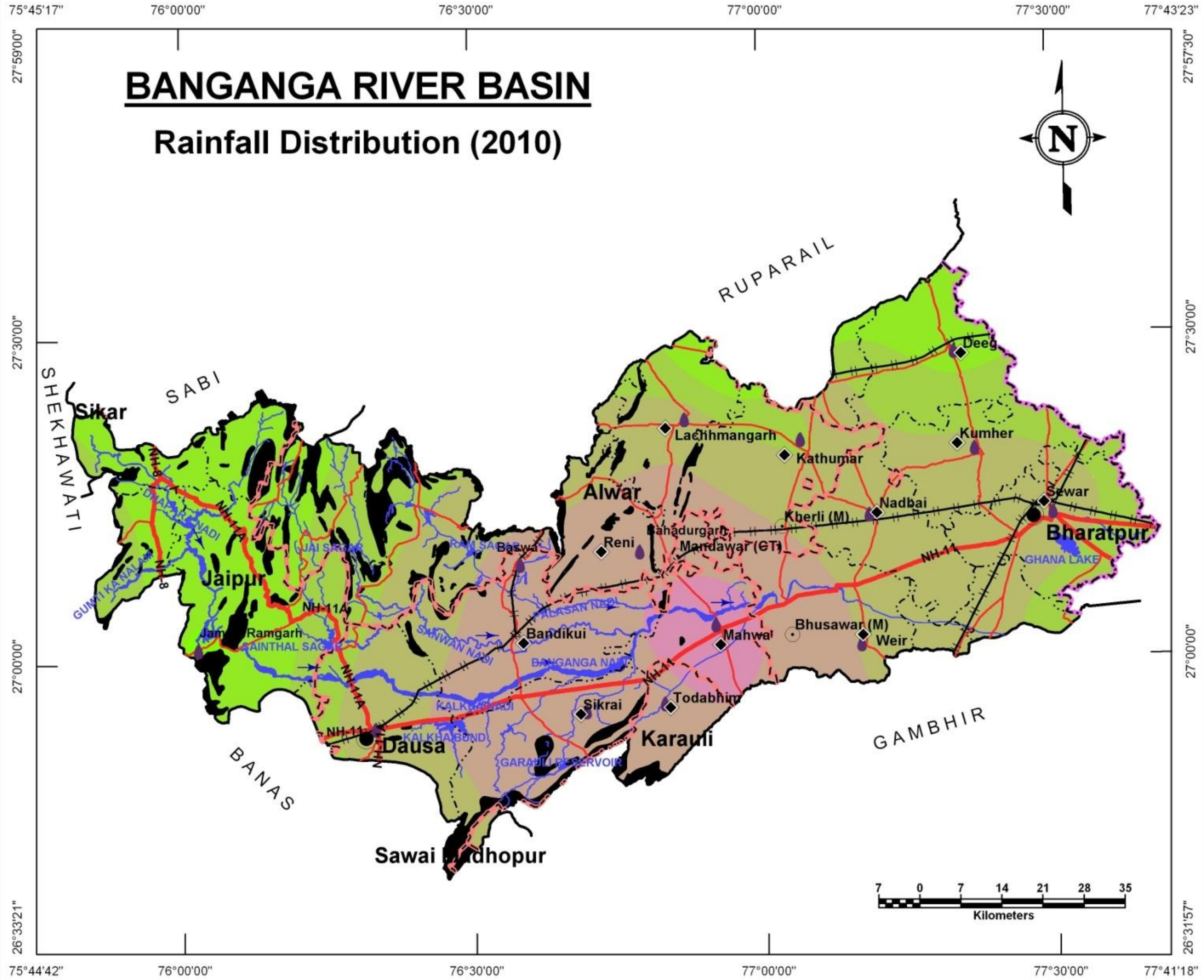
- River / Streams
- Ponds / Reservoirs

Elevation(m amsl):

735 m
625 m
500 m
375 m
250 m
157 m

Source : SRTM DEM

PLATE - III



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- Raingauge Stations
- State Boundary
- District Boundary
- Block Boundary

River Basin Boundary

Roads:

- National Highway
- State Highway

Railway:

- Metre Gauge

Water Bodies:

- River / Streams
- Ponds / Reservoirs

Hills

Isohyets (mm):

- 500-600
- 600-700
- 700-800
- 800-900
- 900-1000
- > 1000

GEOLOGY

The rocks ranging in age between Archaean Granites and Gneisses to Quaternary alluvium are found in the basin.

Pre-Delhi Group: This group is largely represented by pink to grey granites and gneisses. These are compact, jointed and foliated in nature. Weathering in these rocks has been noticed to various depths. Delhi Super Group of rocks un-conformably (Eparchean unconformity) overlay the Pre-Delhi Group of rocks.

Delhi Super Group: This Super Group is divided into two Groups i.e., Alwar and Ajabgarh Groups separated by Kushalgarh limestone and Hornstone breccia formation. The Alwars are predominantly arenaceous whereas the Ajabgarhs are argillaceous in nature appearing as long linear ridges prominently appearing distinctly above the surrounding alluvial tract.

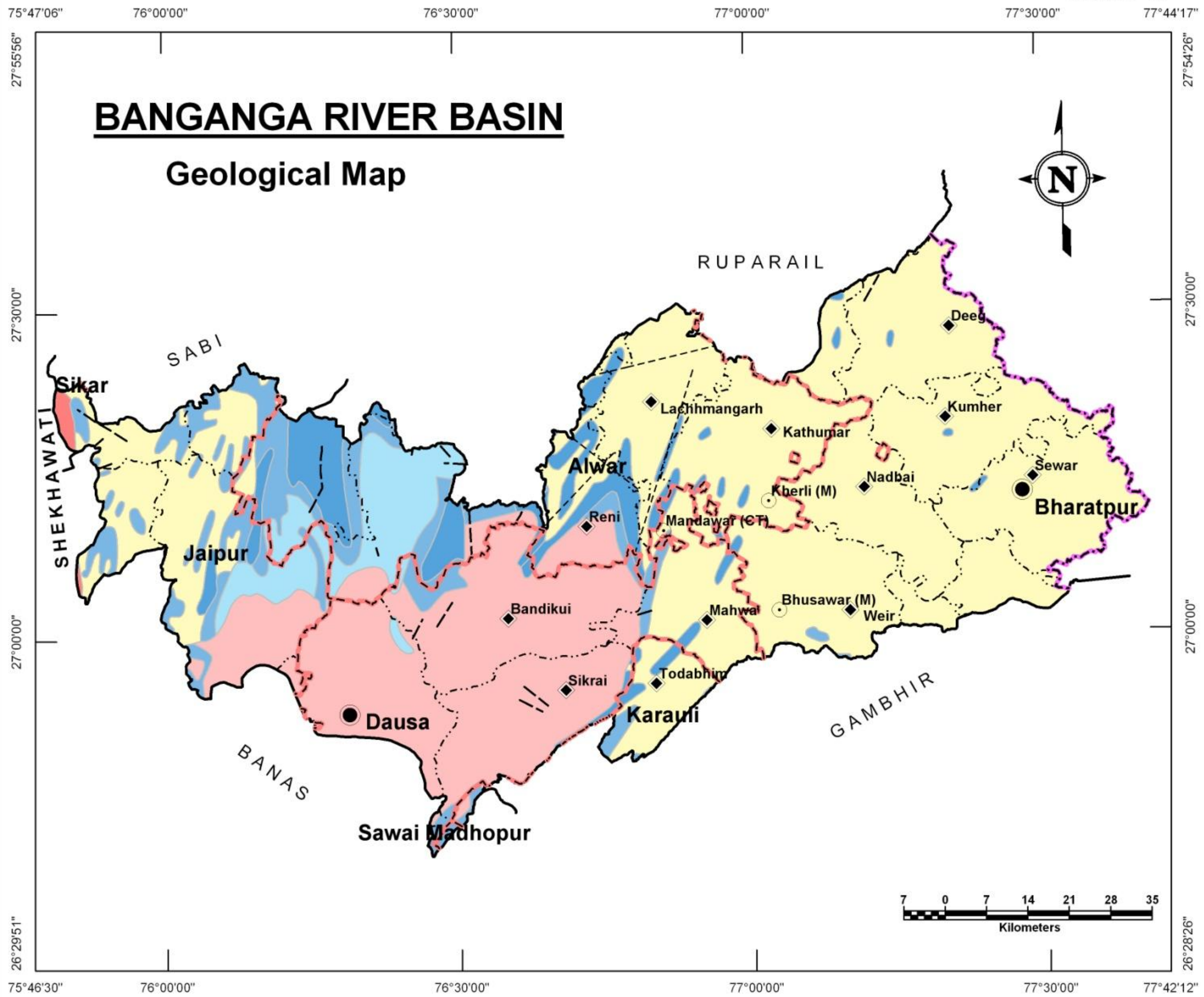
Quaternary Alluvium: River deposited alluvium constituted of sand, silt clay and kankar constitutes the Quaternary Alluvium. At places, wind-blown sand mixes with alluvial sediments as top layer and also often occurs as stabilized sand dunes.

Super Group	Group	Formation
	Quaternary	Alluvium and wind-blown sand comprised of sand, silt, gravel and talus, clay and kankar.
Delhi	Ajabgarh	Argillaceous meta sediments, consisting of phyllities, quartzites, amphibolities (and soapstone), impure dolomitic limestone, Kushalgarh limestone and (tectonic) hornstone breccia.
	Alwar	Arenaceous meta sediments, consisting of quartzites, conglomerates, gritty quartzites interbedded with phyllite, quartz sericite schist, etc.
-----------------x-----------------x-----------------x---------------- x-----------------UNCONFORMITY----------------x------------------x-------------------x-----------------		
Bhilwara and Aravali (pre-Delhi), Granites, Granitic Gneisses (Archean)		

GEOMORPHOLOGY

Origin	Landform Unit	Description
Aeolian	Eolian Plain	Formed by eolian activity, with sand dunes of varying height, size, slope. Long stretches of sand sheet. Gently sloping flat to undulating plain, comprised of fine to medium grained sand and silt. Also scattered xerophytic vegetation.
	Sandy Plain	Formed of aeolian activity, wind-blown sand with gentle sloping to undulating plain, comprising of coarse sand, fine sand, silt & clay.
Denudational	Pediment	Broad gently sloping rock flooring, erosional surface of low relief between hill and plain, comprised of varied lithology, criss-crossed by fractures & faults.
	Intermontane Valley	Depression between mountains, generally broad & linear, filled with colluvial deposits.
Fluvial	Alluvial Plain	Mainly undulating landscape formed due to fluvial activity, comprising of gravels, sand, silt and clay. Terrain mainly undulating, produced by extensive deposition of alluvium.
	Alluvial Plain (Sandy)	Flat to gentle undulating plain formed due to fluvial activity, mainly consists of gravels, sand, silt and clay with unconsolidated material of varying lithology, predominantly sand along river.
	Valley Fill	Formed by fluvial activity, usually at lower topographic locations, comprising of boulders, cobbles, pebbles, gravels, sand, silt and clay. The unit has consolidated sediment deposits.
	Ravine	Small, narrow, deep, depression, smaller than gorges, larger than gulley, usually carved by running water.
	Flood Plain	The surface or strip of relatively smooth land adjacent to a river channel formed by river and covered with water when river over flows its bank. Normally subject to periodic flooding.
	Water logged/ Wetland	Area submerged in water or area having very shallow water table. So that it submerges in water during rainy season.
Hills	Structural Hill	Linear to arcuate hills showing definite trend-lines with varying lithology associated with folding, faulting etc.
Structural	Dissected Plateau	Plateau, criss-crossed by fractures forming deep valleys.

PLATE - IV



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

River Basin Boundary

Structural Features:

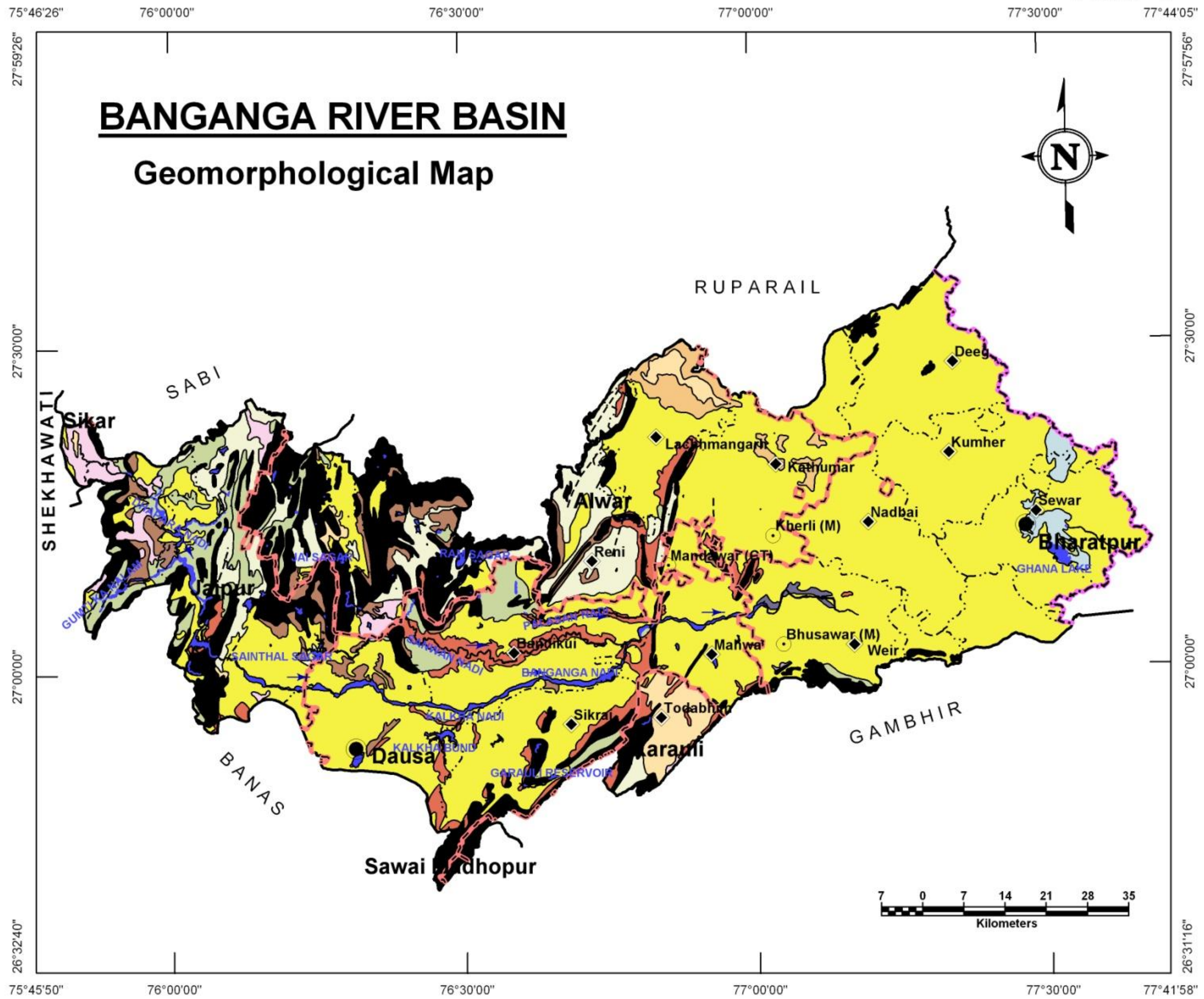
- Fault
- Lineament

Geology:

- Alluvium and wind blown sand
- Ajabgarh Group
- Alwar Group
- Raialo Group
- Mangalwar Complex
- Sand Mata Complex

Source : District Resource Map of Rajasthan - GSI

PLATE - V



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

River Basin Boundary

Water Bodies:

- River/Ponds / Reservoirs

Structural Features:

- Fault/Fractures/Lineament

Hills:

- Structural/Denudational/ Linear Ridge

Landform Units:

Fluvial Origin:

- Alluvial plain
- Alluvial Plain (Sandy)
- Ravine
- Flood Plain
- Valley Fill
- Waterlogged Area

Denudational Origin:

- Burried Pediment
- Pediment
- Intermontane Valley

Aeolian Origin:

- Sandy Plain
- Eolian Plain

Structural Origin:

- Dissected Plateau

Source : Ground Water Atlas of Rajasthan - SRSAC & GWD, Rajasthan

AQUIFERS

The structural lineaments and general strike of ridges is parallel to the Great (strike-slip) Boundary Fault, of NE-SW to NNE-SSW alignment of the Aravali ranges. In the immediate vicinity of these ridges there is, generally, a coarse grained belt (talus, fault debris) having favorable ground water prospects.

In the Delhi rocks, mainly the quartzites, as well as in the pre-Delhi rocks, ground water occurs in the weathered zones of granites and gneisses and in secondary openings, and these comprise the only, rather poor, exploitable aquifers.

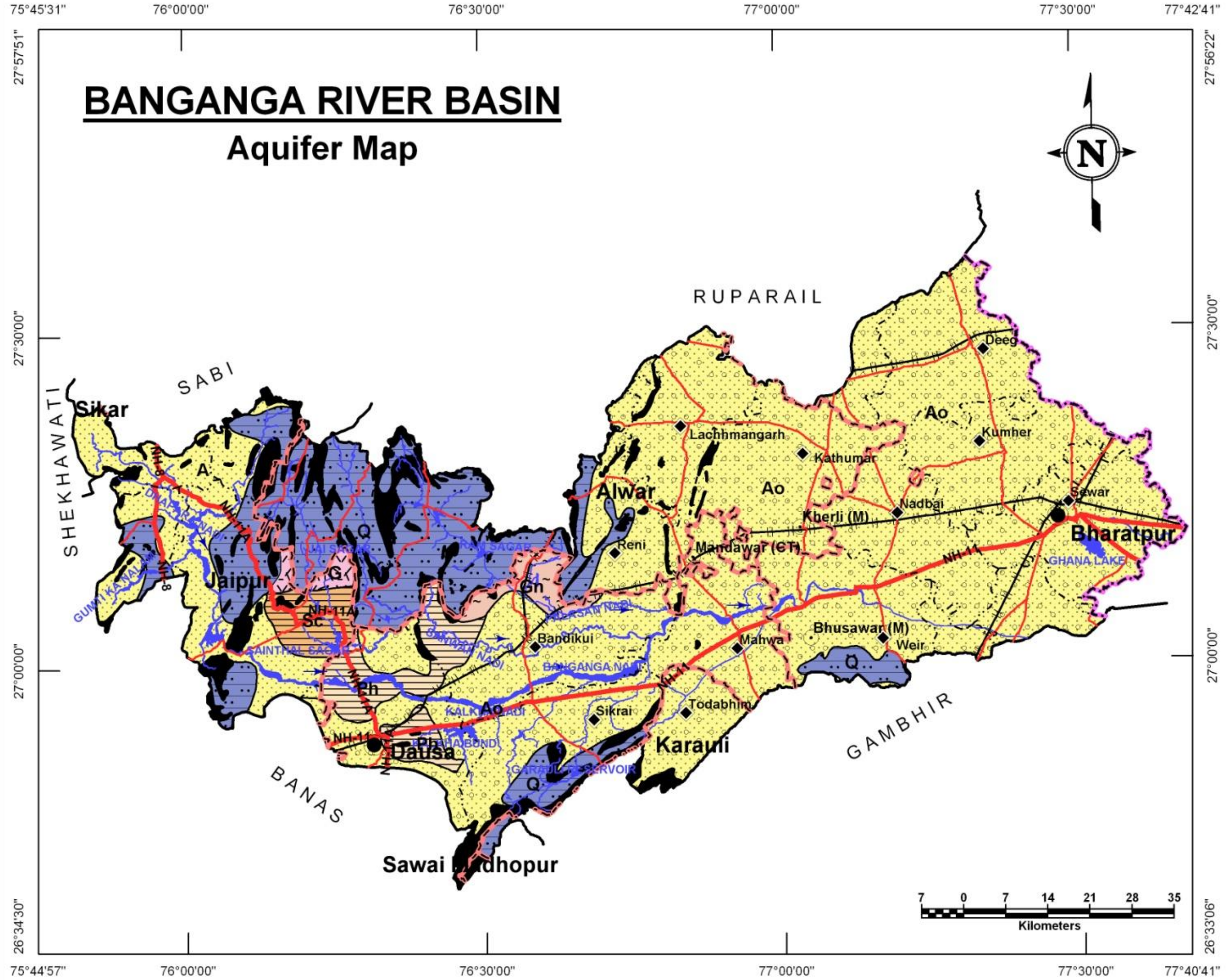
Aquifer in Potential Zone	Area (sq km)	Description of the unit/Occurrence
Younger Alluvium	1,254.7	Alluvium generally comprising gravel, sand, silt, sandy and silty clay, clay and clay kankar. Gravel generally occurs as regular bed over the bedrock, although it may also be present in dispersed form or as thin lenses embedded in sand horizons. The maximum thickness of alluvium is about 225m at Bahaj of Bharatpur district. Sandy clay, sand and kankar generally form the water table aquifer, which is normally tapped by dug wells.
Older Alluvium	4,972.6	
Quartzite	1,667.0	Groundwater occurs in weathered and fractured zones of quartzite. The yield of dug well in Quartzite ranges between 40.00 to 75.00 cum/day, whereas in tube well varies from 0.40 to 9.00cum/hr and upto the maximum yield of 27.00cum/hr at Boronda of Dausa District.
Schist	136.4	Here also water occurs in weathered and fractured zones. The yield of dug wells in this formation varies from 3.80 to 43.40m^3/day whereas in tube wells, the discharge varies from 11.00m^3/hr to 25.00m^3/hr.
Phyllite	73.7	Average yield of dug well in this formation is 40.00m^3/day and the discharge of the tube well is about 18.00m^3/hr.
Granite	49.0	Very limited in occurrence, and wherever the rock is fractured or jointed, fresh groundwater is generally found.
Gneiss	105.1	The yield of dugwell in Gneiss varies from 35.00 to 45.00cum/day while in Piezometer constructed at Baswa, District Dausa, the discharge observed as 6.75cum/hr.
Non Potential Zone (Hills)	530.3	Areas occupied by hills
Total	**8,788.7**	

LOCATION OF GROUNDWATER MONITORING WELLS

The basin has large number of groundwater monitoring stations (260) in the basin owned by RGWD (202) and CGWB (58). Benchmarking and optimization studies recommend that for optimal monitoring of the aquifers in the basin, 52 additional wells for water level monitoring and 160 additional wells for water quality monitoring be added to existing network

District Name	Existing Ground Water Monitoring Stations			Recommended Additional Ground Water Monitoring Stations	
	CGWB	RGWD	Total	Water Level	Water quality
Alwar	11	27	38	30	52
Bharatpur	25	95	120	-	36
Dausa	21	54	75	11	60
Jaipur	-	18	18	11	3
Karauli	1	8	9	-	9
Sawai Madhopur	-	-	-	-	-
Grand Total	**58**	**202**	**260**	**52**	**160**

PLATE - VI



LEGEND

Admin Boundary:

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- Block Boundary

River Basin Boundary

Roads:

- National Highway
- State Highway

Railway:

- Metre Gauge

Water Bodies:

- River / Streams
- Ponds / Reservoirs

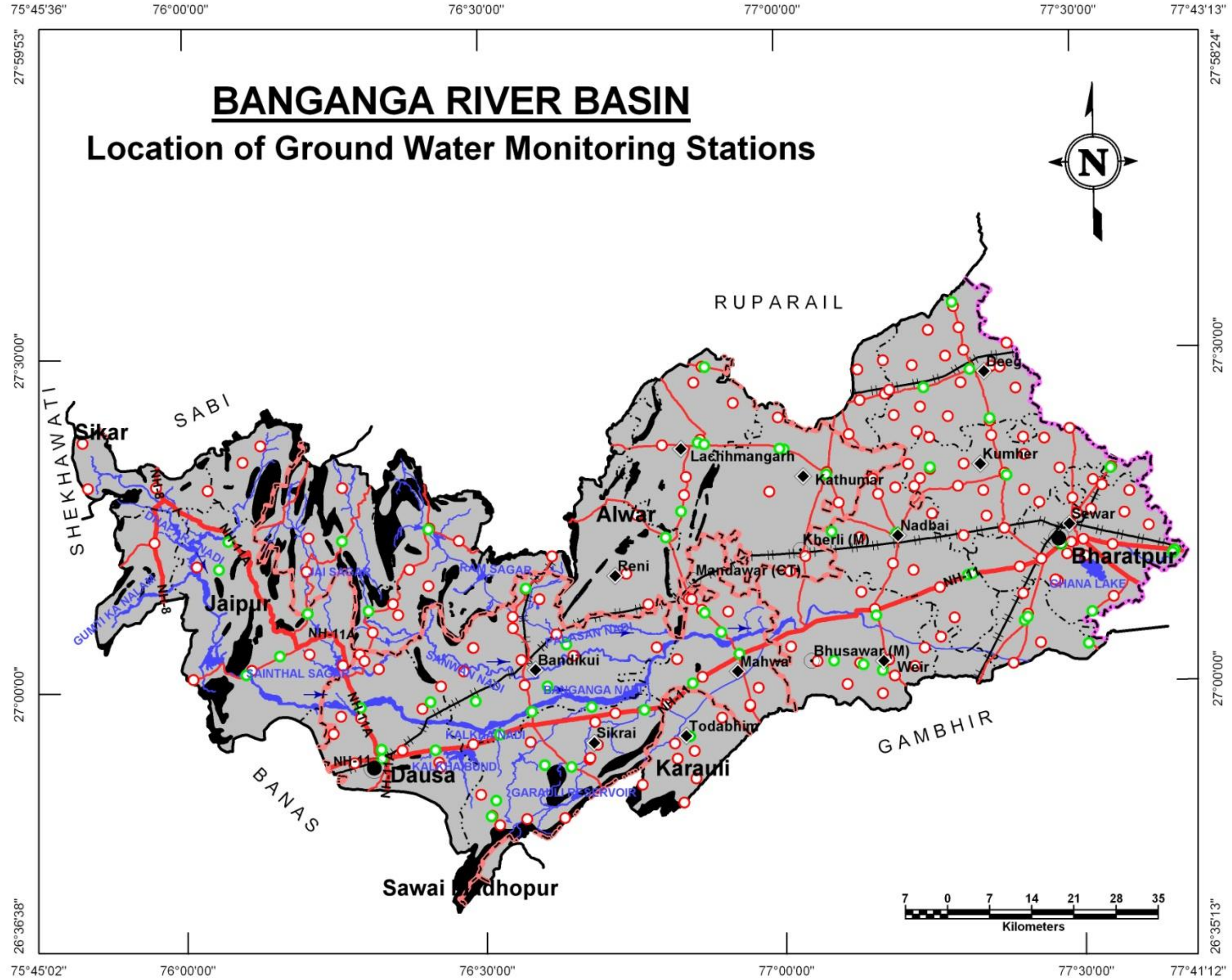
Hills

Aquifer:

- Younger Alluvium
- Older Alluvium
- Phyllite
- Schist
- Quartzite
- Granite
- Gneiss

Source: Ground Water Potential Zone Map - GWD,Rajasthan

PLATE - VII



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- Block Boundary

River Basin Boundary

Roads:

- National Highway
- State Highway

Railway:

- Metre Gauge

Water Bodies:

- River / Streams
- Ponds / Reservoirs

Hills

Ground Water Monitoring Stations:

- CGWB
- RGWD

LOCATION OF EXPLORATORY WELLS

260 exploratory wells are present in the basin drilled in the past by RGWD (202) and CGWB (58) which form the basis for aquifer delineation and three dimensional visualization of the same.

District Name	Exploratory Wells		
	CGWB	RGWD	Total
Alwar	11	27	38
Bharatpur	25	95	120
Dausa	21	54	75
Jaipur	-	18	18
Karauli	1	8	9
Sawai Madhopur	-	-	-
Grand Total	**58**	**202**	**260**

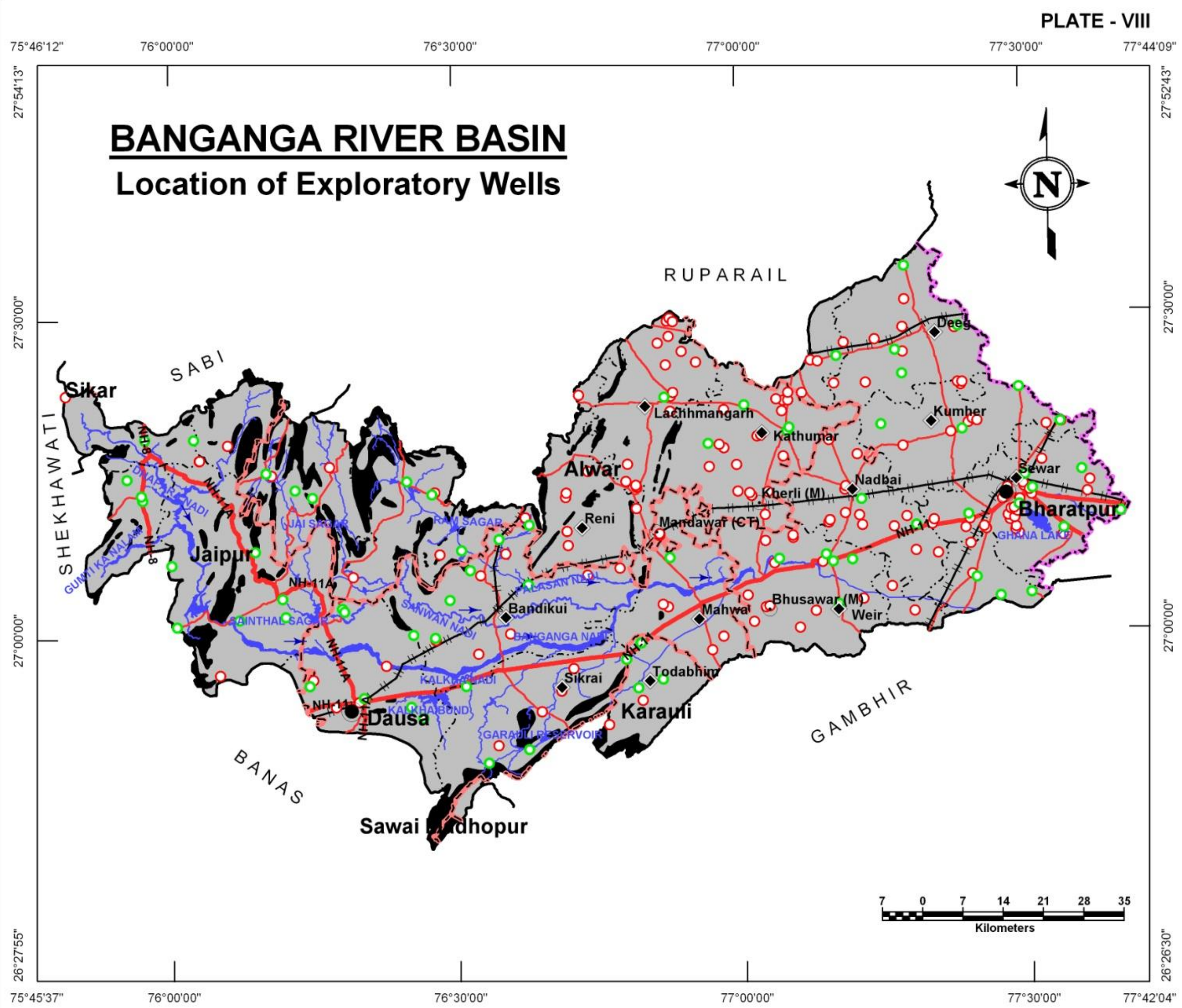
DEPTH TO WATER LEVEL (PRE MONSOON – 2010)

The general depth to water level in the basin ranges from 10 to 30 meters all over the alluvial areas except for the locations north of Bandikui, Karauli and Amber where deeper water levels of around 30 to 50m have been observed.

Table: District wise area covered in each water level depth range

Depth to water level (m bgl) Pre Monsoon - 2010	District wise area (sq km)*							Total Area (sq km)
	Alwar	Bharatpur	Dausa	Jaipur	Karauli	Sawai Madhopur	Sikar	
< 10	19.8	1,056.8	-	-	-	-	-	**1,076.6**
10-20	1,010.1	1,382.4	339.8	169.8	77.3	-	-	**2,979.3**
20-30	808.3	289.0	1,007.2	645.4	70.0	-	-	**2,819.8**
30-40	240.7	18.7	648.2	242.9	34.2	-	-	**1,184.7**
40-50	0.1	-	32.9	118.9	22.2	-	0.4	**174.4**
> 50	-	-	-	-	21.2	-	-	**21.2**
Total	**2,079.0**	**2,746.8**	**2,028.1**	**1,176.9**	**224.9**	**-**	**0.4**	**8,256.0**

* The area covered in the derived maps is less than the total basin area since the hills have been excluded from interpolation/contouring.



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River Basin Boundary

Roads:

- National Highway
- State Highway

Railway:

- Metre Gauge

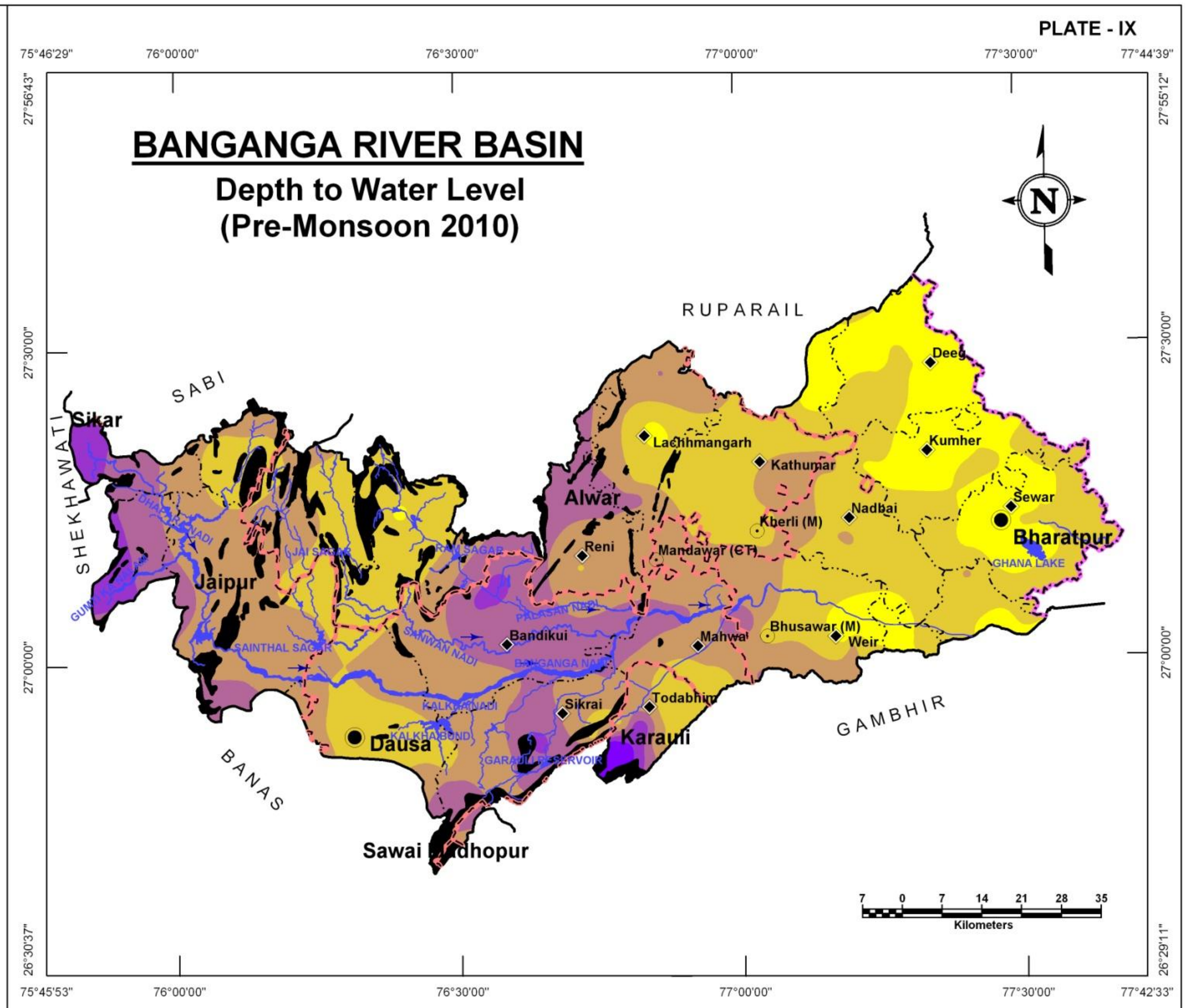
Water Bodies:

- River / Streams
- Ponds / Reservoirs

Hills

Exploratory Wells:

- CGWB
- RGWD



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

River Basin Boundary

Water Bodies:

- River / Streams
- Ponds / Reservoirs

Hills

Depth to Water Level (m bgl):

- < 10
- 10-20
- 20-30
- 30-40
- 40-50
- > 50

WATER TABLE ELEVATION (PRE MONSOON – 2010) BANGANGA RIVER BASIN

Very high variation in groundwater elevation has been observed in the basin which ranges from 160m amsl to more than 460m amsl. In the hilly areas in the western part the water table is not only high but also has a steeper gradient. The general groundwater flow direction is towards eastwards. Further in the east where groundwater occurs in alluvial sand, the flow gradients are relatively flatter. In far eastern part, the water table intersects ground and water logging is observed near Ghana Lake.

District Name	District wise Area (sq km) per water table elevation range (m)																												Total Area (sq km)
	160-170	170-180	180-190	190-200	200-210	210-220	220-230	230-240	240-250	250-260	260-270	270-280	280-290	290-300	300-310	310-320	320-330	330-340	340-350	350-360	360-370	370-380	380-390	390-400	400-410	410-420	420-430	430-460	
Alwar	-	-	90	191.8	314.3	336	149.6	137.6	138.6	18.3	15.7	22	32.9	25	52.2	41.9	42.1	44.3	48.9	34.9	43.1	50.1	50.4	48.7	67.1	51.1	30.6	2	**2,079.0**
Bharatpur	383.9	1,033.40	685.4	413.9	165.5	64.7	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	**2,746.8**
Dausa	-	-	-	18	118.3	140.1	122.7	162.7	401.5	163.7	129.7	125.4	115.8	147.5	281.7	101	0	-	-	-	-	-	-	-	-	-	-	-	**2,028.1**
Jaipur	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1.3	36	140.8	89.1	64.4	73.2	62.6	51.9	52.6	102.1	98.3	134.7	111.5	158.5	**1,176.9**
Karauli	-	-	-	0.5	33	30.9	75.8	67.9	0	0	0.2	0.1	11.2	5.3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	**224.9**
Sawai Madhopur	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	**-**
Sikar	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0.4	-	-	-	-	-	-	-	-	**-**
Total	**383.9**	**1,033.40**	**775.4**	**624.2**	**631**	**571.6**	**348.1**	**368.2**	**540**	**181.9**	**145.6**	**147.5**	**159.9**	**177.8**	**335.1**	**179**	**182.9**	**133.3**	**113.4**	**108.5**	**105.7**	**102**	**103**	**150.8**	**165.4**	**185.9**	**142**	**160.5**	**8,256.0**

WATER LEVEL FLUCTUATION (PRE TO POST MONSOON 2010)

The red shaded areas in the Plate XI indicate the areas that showed negative fluctuation i.e., water table has gone down in Post-Monsoon period as compared to Pre-Monsoon water levels. This map is very significant as it indicates the general recharge situation post monsoon season of the year. Except for few pockets around Bandikui, Lakshmangarh and North of Jamwa Ramgarh, the water table has generally risen.

District Name	District wise area coverage (sq km) within fluctuation range											Total Area (sq km)
	-6 to -4m	-4 to -2m	-2 to 0m	0 to 2m	2 to 4m	4 to 6m	6 to 8m	8 to 10m	10 to 12m	12 to 14m	14 to 16m	
Alwar	-	15.3	206.3	671.2	521.7	465.6	161.4	23.2	10.6	3.6	-	**2,079.0**
Bharatpur	-	-	22.3	1,190.0	1,180.6	250.9	62.5	26.6	13.3	0.6	-	**2,746.8**
Dausa	-	14.6	422.2	786.2	457.0	199.8	43.1	35.9	44.6	18.4	6.1	**2,028.1**
Jaipur	6.1	18.2	37.9	659.3	331.2	75.1	36.5	12.4	0.1	-	-	**1,176.9**
Karauli	-	-	-	182.7	40.4	1.8	-	-	-	-	-	**224.9**
Sawai Madhopur	-	-	-	-	-	-	-	-	-	-	-	**-**
Sikar	-	-	-	0.4	-	-	-	-	-	-	-	**0.4**
Total	**6.1**	**48.2**	**688.8**	**3,489.7**	**2,531.0**	**993.2**	**303.5**	**98.1**	**68.7**	**22.6**	**6.1**	**8,256.0**

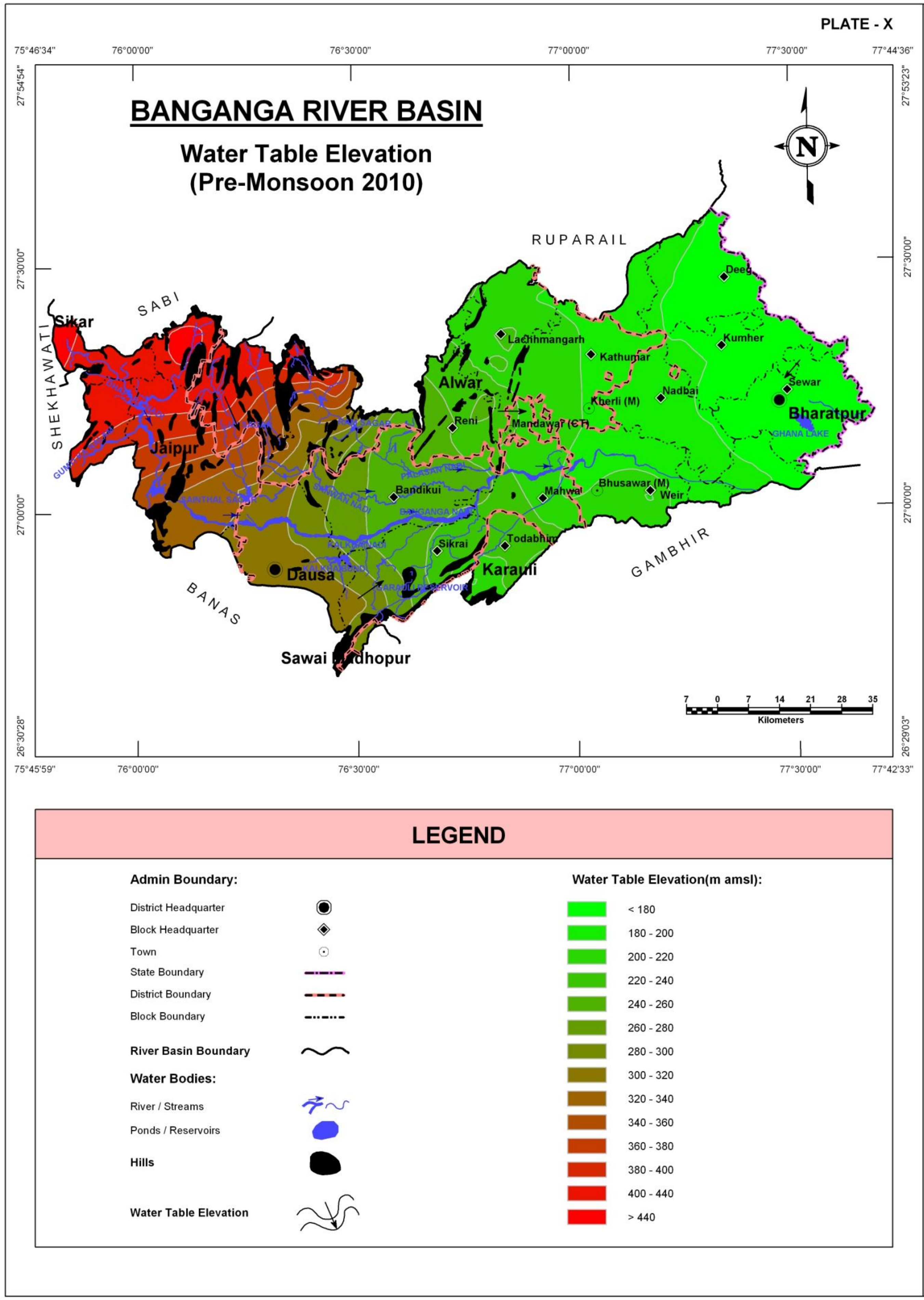


PLATE - X
BANGANGA RIVER BASIN
Water Table Elevation
(Pre-Monsoon 2010)
RUPARAIL
SABI
SHEKHAWATI
BANAS
GAMBHIR
Sikar
Jaipur
Dausa
Sawai Madhopur
Alwar
Lachhmangarh
Kathumar
Reni
Mandawar (CT)
Kherli (M)
Nadbai
Deeg
Kumher
Sewar
Bharatpur
Bandikui
Mahwa
Bhusawar (M)
Weir
Sikrai
Todabhim
Karauli
Kilometers
LEGEND
Admin Boundary:
District Headquarter
Block Headquarter
Town
State Boundary
District Boundary
Block Boundary
River Basin Boundary
Water Bodies:
River / Streams
Ponds / Reservoirs
Hills
Water Table Elevation
Water Table Elevation(m amsl):
< 180
180 - 200
200 - 220
220 - 240
240 - 260
260 - 280
280 - 300
300 - 320
320 - 340
340 - 360
360 - 380
380 - 400
400 - 440
> 440

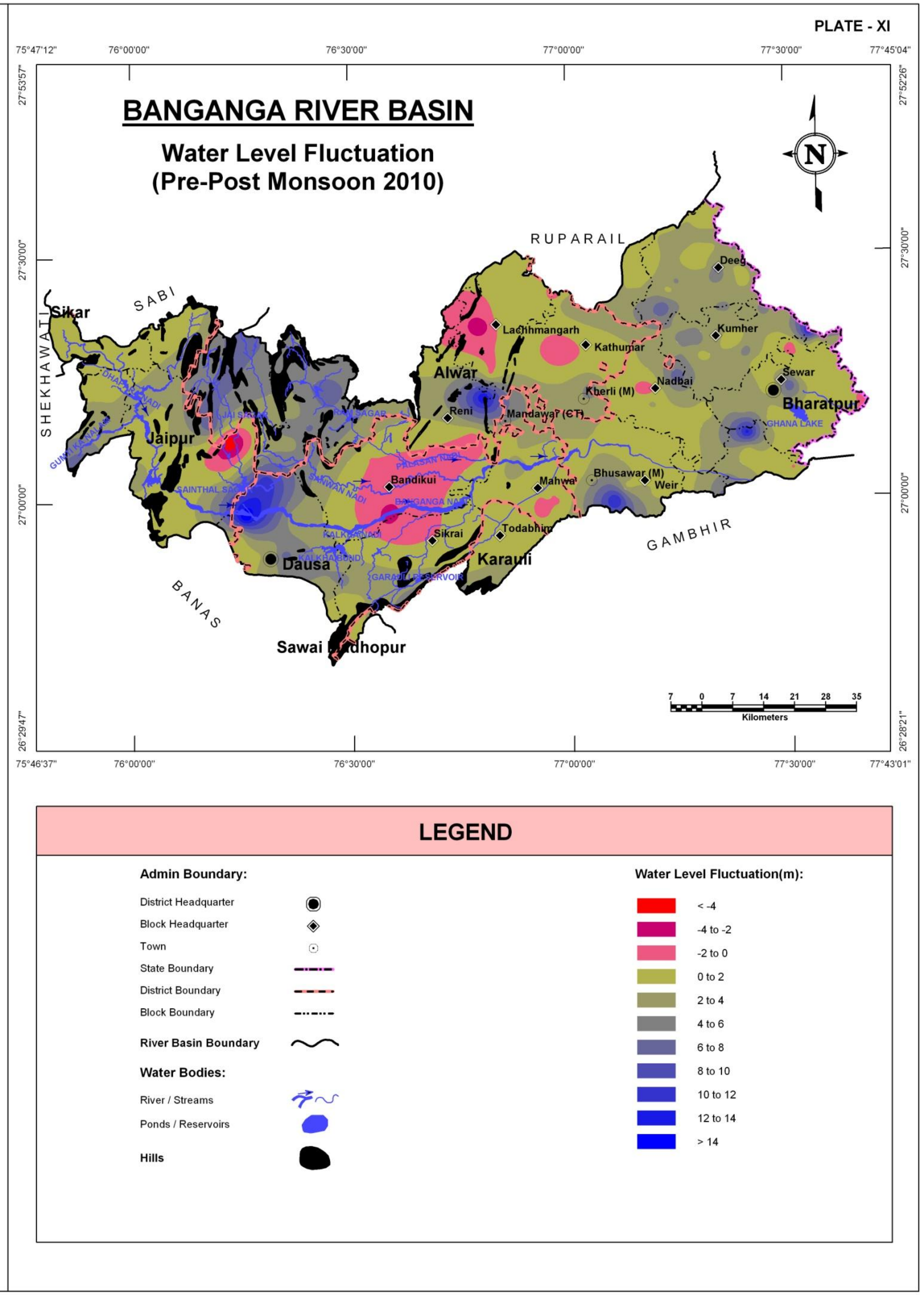


PLATE - XI
BANGANGA RIVER BASIN
Water Level Fluctuation
(Pre-Post Monsoon 2010)
RUPARAIL
SABI
SHEKHAWATI
BANAS
GAMBHIR
Sikar
Jaipur
Dausa
Sawai Madhopur
Alwar
Lachhmangarh
Kathumar
Reni
Mandawar (CT)
Kherli (M)
Nadbai
Deeg
Kumher
Sewar
Bharatpur
Bandikui
Mahwa
Bhusawar (M)
Weir
Sikrai
Todabhim
Karauli
Kilometers
LEGEND
Admin Boundary:
District Headquarter
Block Headquarter
Town
State Boundary
District Boundary
Block Boundary
River Basin Boundary
Water Bodies:
River / Streams
Ponds / Reservoirs
Hills
Water Level Fluctuation(m):
< -4
-4 to -2
-2 to 0
0 to 2
2 to 4
4 to 6
6 to 8
8 to 10
10 to 12
12 to 14
> 14

ELECTRICAL CONDUCTIVITY DISTRIBUTION

BANGANGA RIVER BASIN

The electrical conductivity of ground water ranges from less than 2000 µS/cm to more than 30,000 µS/cm in the basin. The areas marked in red colour are high salinity areas with EC values more than 4000 µS/cm. Such area are largely found in eastern part of the areas and in alluvial aquifers, as seen around Bharatpur, Nadbai, Kumber, Deeg, Lakshmangarh, and in Dausa. The fresh water areas are found within and around hardrock areas. Rest of the areas has generally higher salt content in groundwater. The analysis is based on average of EC values observed during Pre-Monsoon between years 2005-09.

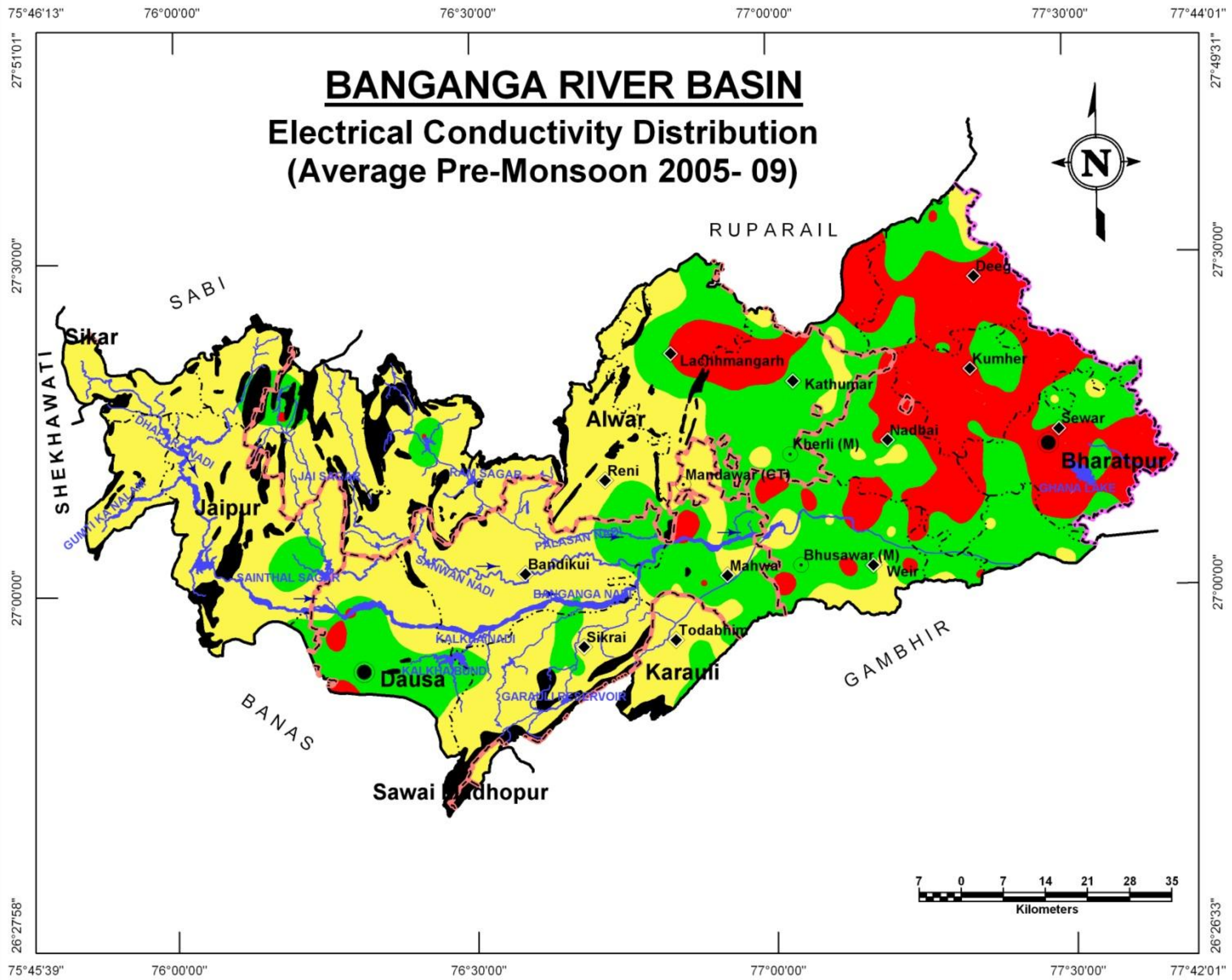
Electrical Conductivity Ranges (µS/cm at 25^{o}C) (Ave. of years 2005-09)	District wise area coverage (sq km)														Total Area (sq km)
	Alwar		Bharatpur		Dausa		Jaipur		Karauli		Sawai Madhopur		Sikar		
	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	
< 2000	1,217.7	58.6	184.6	6.7	1,315.8	64.9	1,087.7	92.4	190.9	84.9	-	0.0	0.4	100.0	**3,997.0**
2000-4000	669.0	32.2	1,267.7	46.2	667.8	32.9	89.0	7.6	34.0	15.1	-	0.0	-	0.0	**2,727.5**
> 4000	192.3	9.3	1,294.5	47.1	44.5	2.2	0.2	0.0	-	0.0	-	0.0	-	0.0	**1,531.5**
Total	**2,079.0**	**100.0**	**2,746.8**	**100.0**	**2,028.1**	**100.0**	**1,176.9**	**100.0**	**224.9**	**100.0**	**-**	**0.0**	**0.4**	**100.0**	**8,256.0**

CHLORIDE DISTRIBUTION

The chloride distribution also follows the same pattern of distribution as that of Electrical conductivity. The analysis is based on average of Chloride values observed during Pre-Monsoon between years 2005-09. High chloride concentration in groundwater is found in the eastern part around Bharatpur, Nadbai, Deeg, Kumher, Kathumar, Lakshmangarh and a small patch around Dausa. Bharatpur district has largest area under high salinity, whereas Dausa, Alwar and Jaipur districts have large areas with freshwater.

Chloride Concentration Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)														Total Area (sq km)
	Alwar		Bharatpur		Dausa		Jaipur		Karauli		Sawai Madhopur		Sikar		
	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	
< 250	1,217.7	58.6	184.6	6.7	1,315.8	64.9	1,087.7	92.4	190.9	84.9	-	0.0	0.4	100.0	**3,997.0**
250-1000	669.0	32.2	1,267.7	46.2	667.8	32.9	89.0	7.6	34.0	15.1	-	0.0	-	0.0	**2,727.5**
> 1000	192.3	9.3	1,294.5	47.1	44.5	2.2	0.2	0.0	-	0.0	-	0.0	-	0.0	**1,531.5**
Total	**2,079.0**	**100.0**	**2,746.8**	**100.0**	**2,028.1**	**100.0**	**1,176.9**	**100.0**	**224.9**	**100.0**	**-**	**0.0**	**0.4**	**100.0**	**8,256.0**

PLATE - XII



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

River Basin Boundary

Water Bodies:

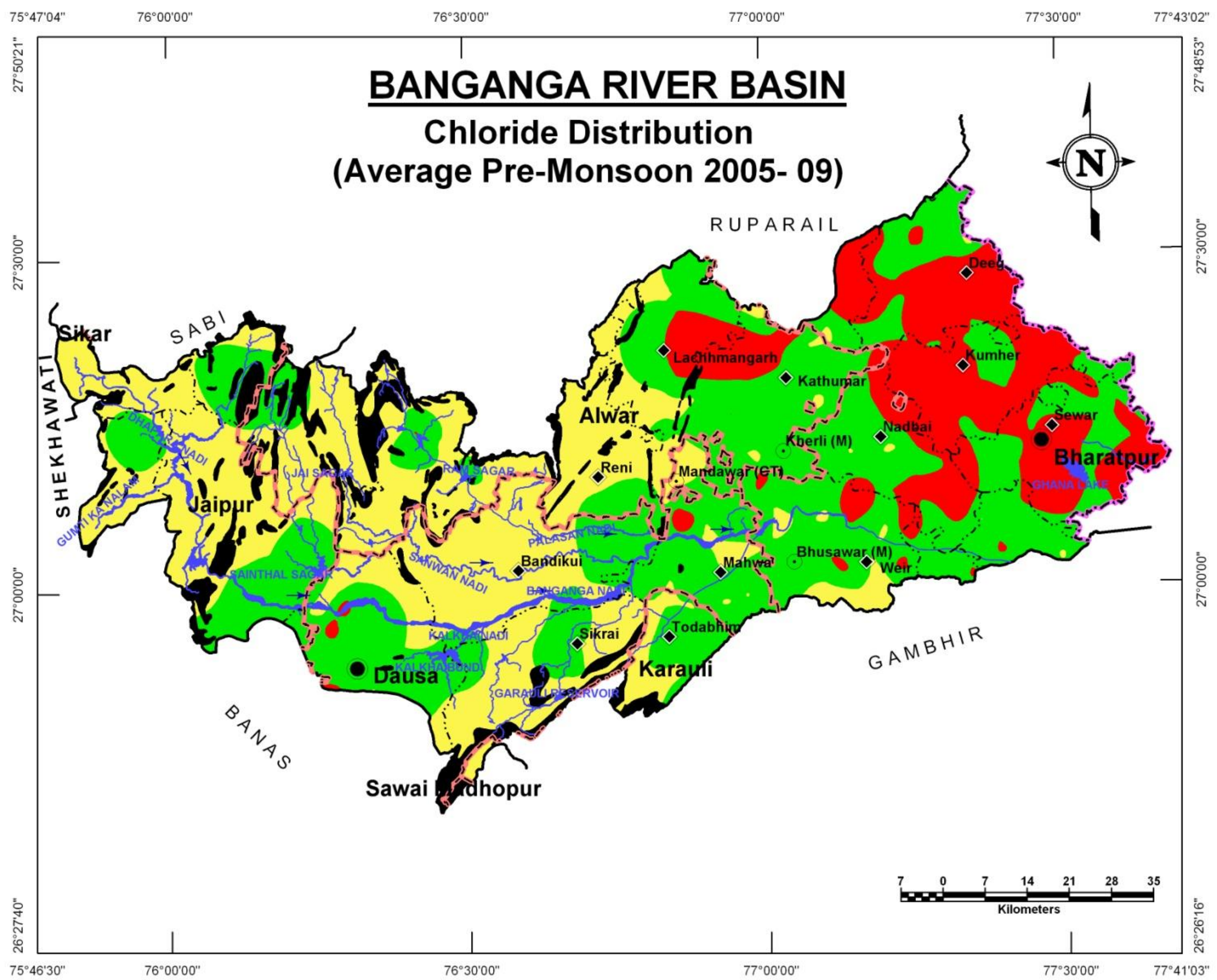
- River / Streams
- Ponds / Reservoirs

Hills

Electrical Conductivity (μS/cm at 25°C):

- < 2000
- 2000-4000
- > 4000

PLATE - XIII



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

River Basin Boundary

Water Bodies:

- River / Streams
- Ponds / Reservoirs

Hills

Chloride Concentration (mg/l):

- < 250
- 250 - 1000
- > 1000

FLUORIDE DISTRIBUTION

High fluoride pockets are seen to the west of Deeg, around Kherli – Mahwa – Weir and to the east of Dausa. In these areas, the concentration is more than 3 mg/l. The general concentration of Fluoride in most of the area is below 1.5 mg/l and some areas between 1.5 to 3 mg/l in between.

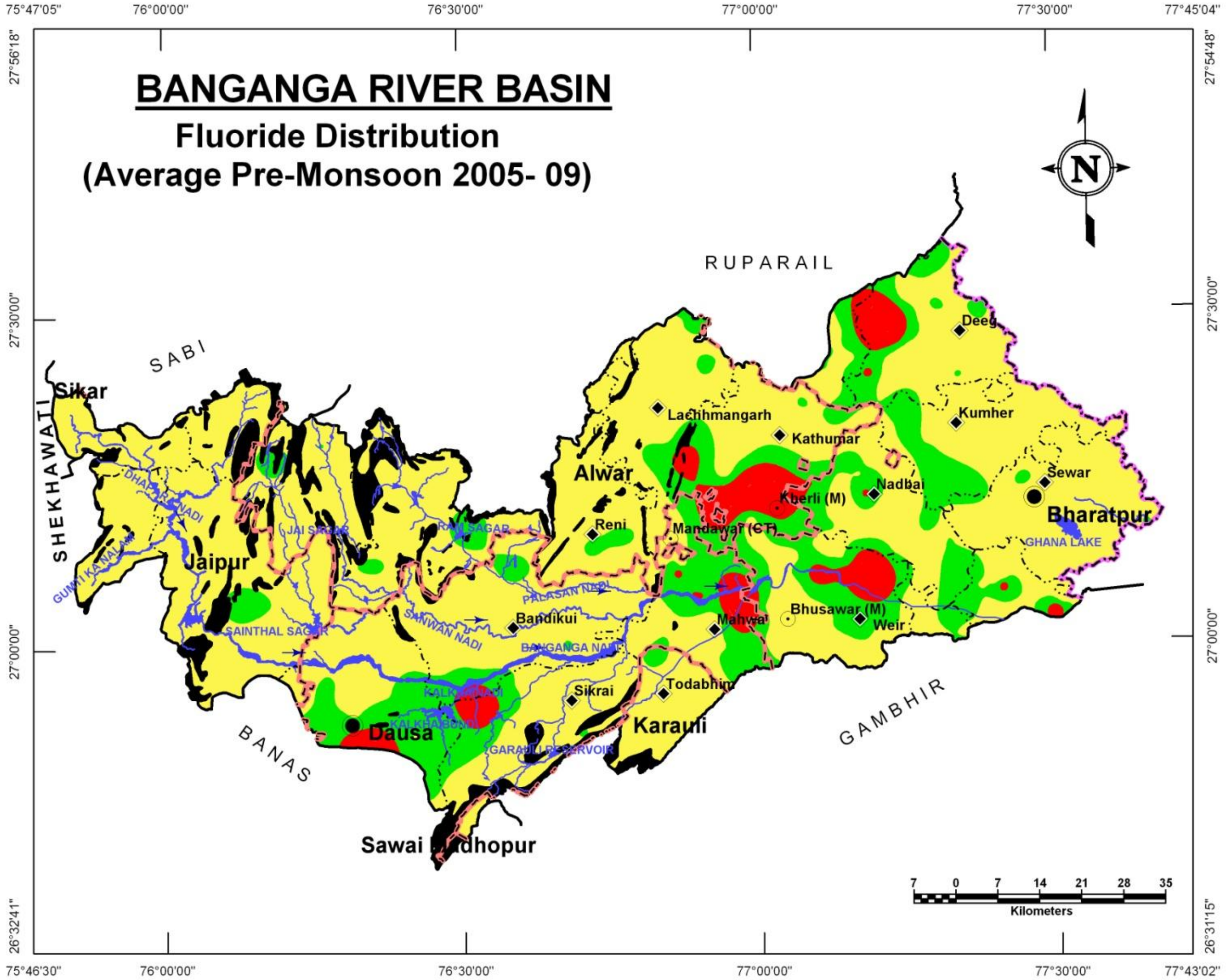
Fluoride Concentration Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)														Total Area (sq km)
	Alwar		Bharatpur		Dausa		Jaipur		Karauli		Sawai Madhopur		Sikar		
	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	
< 1.5	1,217.7	58.6	184.6	6.7	1,315.8	64.9	1,087.7	92.4	190.9	84.9	-	0.0	0.4	100.0	**3,997.0**
1.5-3.0	669.0	32.2	1,267.7	46.2	667.8	32.9	89.0	7.6	34.0	15.1	-	0.0	-	0.0	**2,727.5**
> 3	192.3	9.3	1,294.5	47.1	44.5	2.2	0.2	0.0	-	0.0	-	0.0	-	0.0	**1,531.5**
Total	**2,079.0**	**100.0**	**2,746.8**	**100.0**	**2,028.1**	**100.0**	**1,176.9**	**100.0**	**224.9**	**100.0**	**-**	**0.0**	**0.4**	**100.0**	**8,256.0**

NITRATE DISTRIBUTION

High nitrate concentration in groundwater renders it unsuitable for agriculture purposes. Plate XV shows distribution of Nitrate in groundwater. There are some pockets around Dausa, Rajgarh, Kathumar – Kherli – Bhusawar region that around Deeg have high Nitrate concentration whereas rest of the area has relatively low Nitrate i.e., under 50 mg/l to upto 100 mg/l.

Nitrate Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)														Total Area (sq km)
	Alwar		Bharatpur		Dausa		Jaipur		Karauli		Sawai Madhopur		Sikar		
	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	
<50	1,302.6	62.7	1,294.1	47.1	1,265.4	62.4	759.2	64.5	54.7	24.3	-	0.0	0.4	100.0	**4,676.4**
50-100	583.4	28.1	883.8	32.2	527.9	26.0	344.8	29.3	157.9	70.2	-	0.0	-	0.0	**2,497.9**
>100	192.9	9.3	568.8	20.7	234.8	11.6	72.9	6.2	12.3	5.5	-	0.0	-	0.0	**1,081.7**
Total	**2,079.0**	**100.0**	**2,746.8**	**100.0**	**2,028.1**	**100.0**	**1,176.9**	**100.0**	**224.9**	**100.0**	**-**	**0.0**	**0.4**	**100.0**	**8,256.0**

PLATE - XIV



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

River Basin Boundary

Water Bodies:

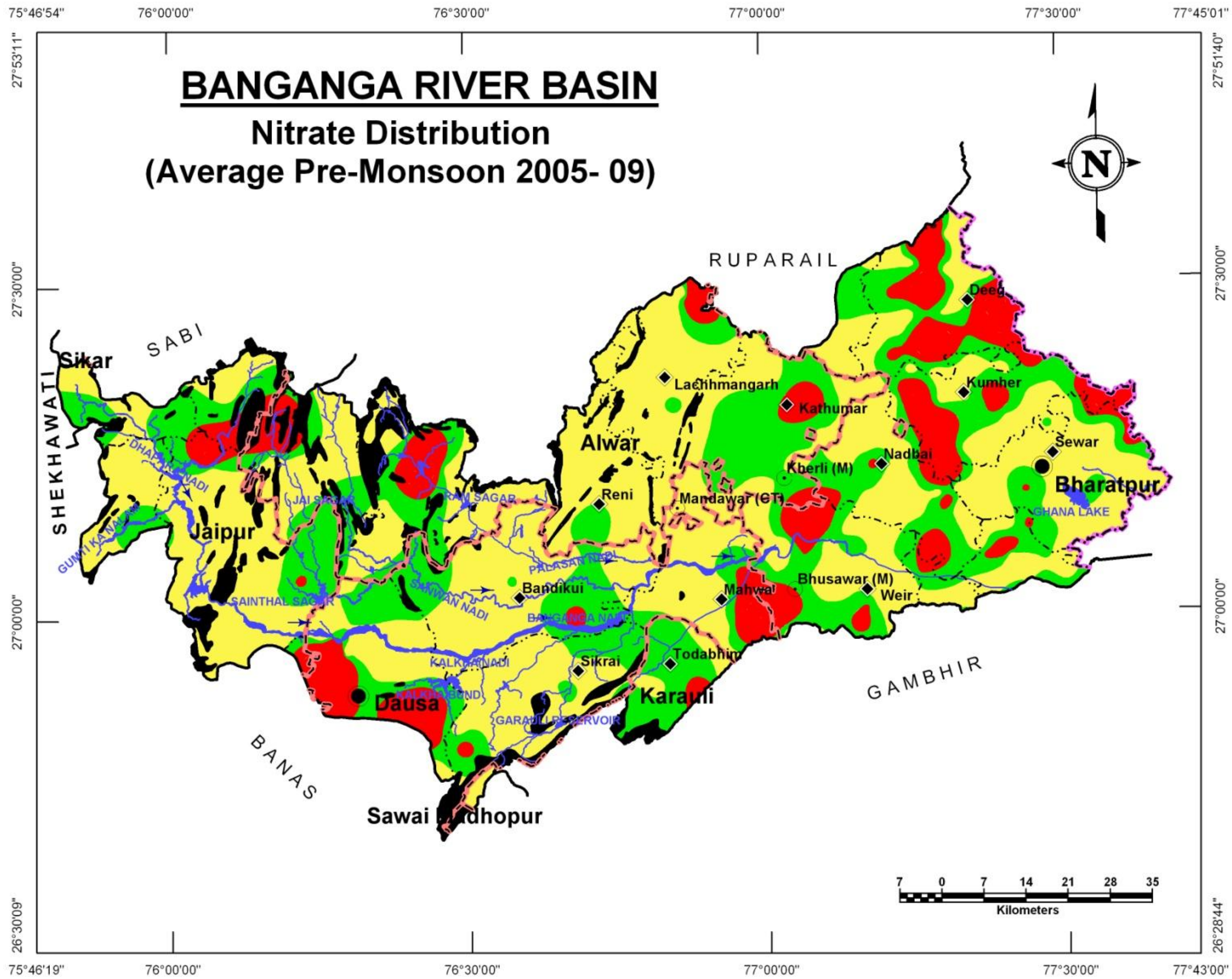
- River / Streams
- Ponds / Reservoirs

Hills

Fluoride Concentration (mg/l):

- < 1.5
- 1.5-3.0
- > 3.0

PLATE - XV



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

River Basin Boundary

Water Bodies:

- River / Streams
- Ponds / Reservoirs

Hills

Nitrate Concentration (mg/l):

- < 50
- 50-100
- > 100

DEPTH TO BEDROCK

The entire area of the Banganga river basin is underlined by the hard rocks at different depths. The major rocks types occurring in the area are Limestone, Shale as sedimentary rocks and Slate, Schist, Phyllite, Quartzite and Gneiss as Metamorphic rocks type. These rocks are overlain by alluvial deposits of sand, clay, silt, kankar, gravels and admixture of all. The depth to bed rock defines the sub surface topography of the occurrence of hard rock beneath alluvial deposits and fractured bedrock. Plate – XVI presents the sub-surface occurrence of bedrock in meters below ground level. It is apparent that the bedrock is at deeper levels in eastern part of the river basin which gradually occurs at shallow depths towards west where even it appears in the form of high hills. The depth to bedrock in the basin varies from less than 20 m bgl in the west and southwest to more than 279 m bgl in the east.

Depth To Bedrock (m bgl)	District Area coverage (sq km)							Total Area (sq km)
	Alwar	Bharatpur	Dausa	Jaipur	Karauli	Sawai Madhopur	Sikar	
20-40	-	-	111.0	37.3	16.8	-	-	**165.1**
40-60	404.4	-	1,238.3	1,102.0	81.5	-	0.4	**2,826.7**
60-80	752.4	1,128.9	594.0	37.6	126.6	-	-	**2,639.4**
80-100	817.1	1,560.6	84.8	-	-	-	-	**2,462.6**
100-120	105.0	57.3	-	-	-	-	-	**162.3**
Total	**2,079.0**	**2,746.8**	**2,028.1**	**1,176.9**	**224.9**	**-**	**0.4**	**8,256.0**

UNCONFINED AQUIFER

Hydrogeological properties are different for alluvial and hard rock aquifers and therefore, this aquifer has been mapped as two separate regions viz, unconfined aquifers in alluvial and in hard rock areas.

Alluvial areas

Except for hardrock exposure areas, this aquifer is present all over the basin with thicknesses varying from very high (reaching 70m) in eastern part in Sewar-Kumher-Dig region to a general less than 30 meter in rest of the basin area. Significant thickness of 30-50m can also be noticed around Weir and Lachhmangarh-Kathumar region.

Hardrock areas

Weathered, fractured and jointed rock formations form the phreatic aquifer in the areas where hardrocks are exposed or are very shallow. These aquifers are largely found in the western part of the basin in the region between Viratnagar, Jamwa Ramgarh, Dausa, Bandikui and Reni. The weathered zone reaches a thickness of more than 40m. In addition to this large region of aquifers formed in weathered and fractured zone, isolated pockets of such unconfined aquifers are found in east of Dausa, Lalsot-Todabhim region and south of Weir where the pocket east of Dausa has shown the presence of >30m of weathered/fractured zone while other two areas have upto 20m of weathered zone.

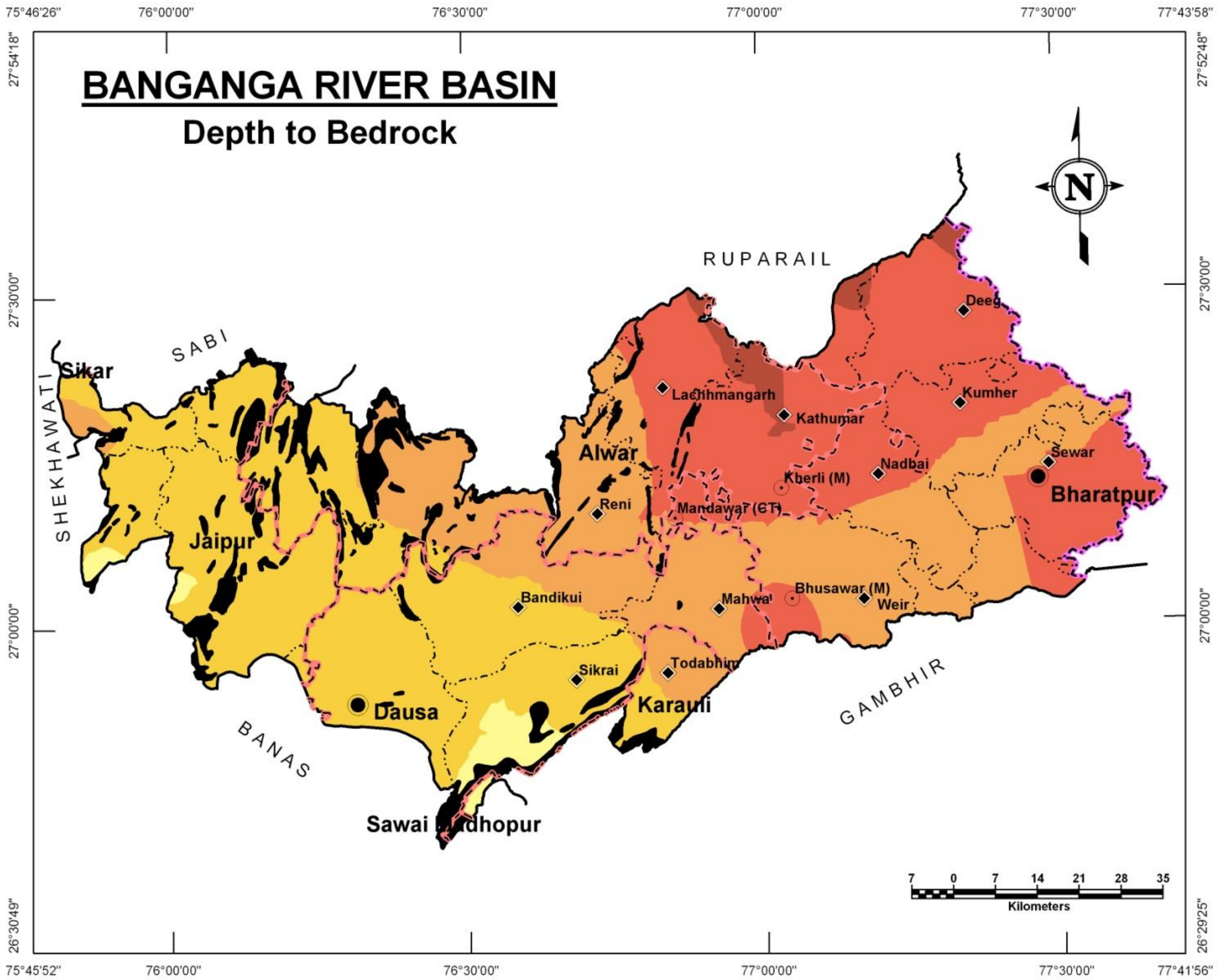
Alluvial areas:

Unconfined aquifer Thickness (m)	District area coverage (sq km)							Total Area (sq km)
	Alwar	Bharatpur	Dausa	Jaipur	Karauli	S. Madhopur	Sikar	
< 10	1,086.6	780.5	1,718.1	852.4	116.7	-	0.4	**4,554.7**
10-20	707.2	671.1	257.5	287.8	57.7	-	-	**1,981.3**
20-30	177.2	663.8	44.4	25.4	48.3	-	-	**959.1**
30-40	95.0	297.5	8.1	-	2.2	-	-	**402.8**
40-50	12.8	178.5	-	-	-	-	-	**191.3**
50-60	-	107.8	-	-	-	-	-	**107.8**
60-70	-	44.4	-	-	-	-	-	**44.4**
70-80	-	3.3	-	-	-	-	-	**3.3**
Total	**2,079.0**	**2,746.8**	**2,028.1**	**1,165.6**	**224.9**	**-**	**0.4**	**8,244.7**

Hardrock areas:

Unconfined aquifer Thickness (m)	District area coverage (sq km)							Total Area (sq km)
	Alwar	Bharatpur	Dausa	Jaipur	Karauli	S. Madhopur	Sikar	
< 10	556.36	72.65	329.04	343.87	16.90	-	-	**1,318.8**
10-20	252.38	0.60	114.86	100.20	-	-	-	**468.0**
20-30	-	-	38.96	25.38	-	-	-	**64.3**
30-40	-	-	8.08	-	-	-	-	**8.1**
Total	**808.7**	**73.3**	**490.9**	**469.5**	**16.9**	**-**	**-**	**1,859.3**

PLATE - XVI



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

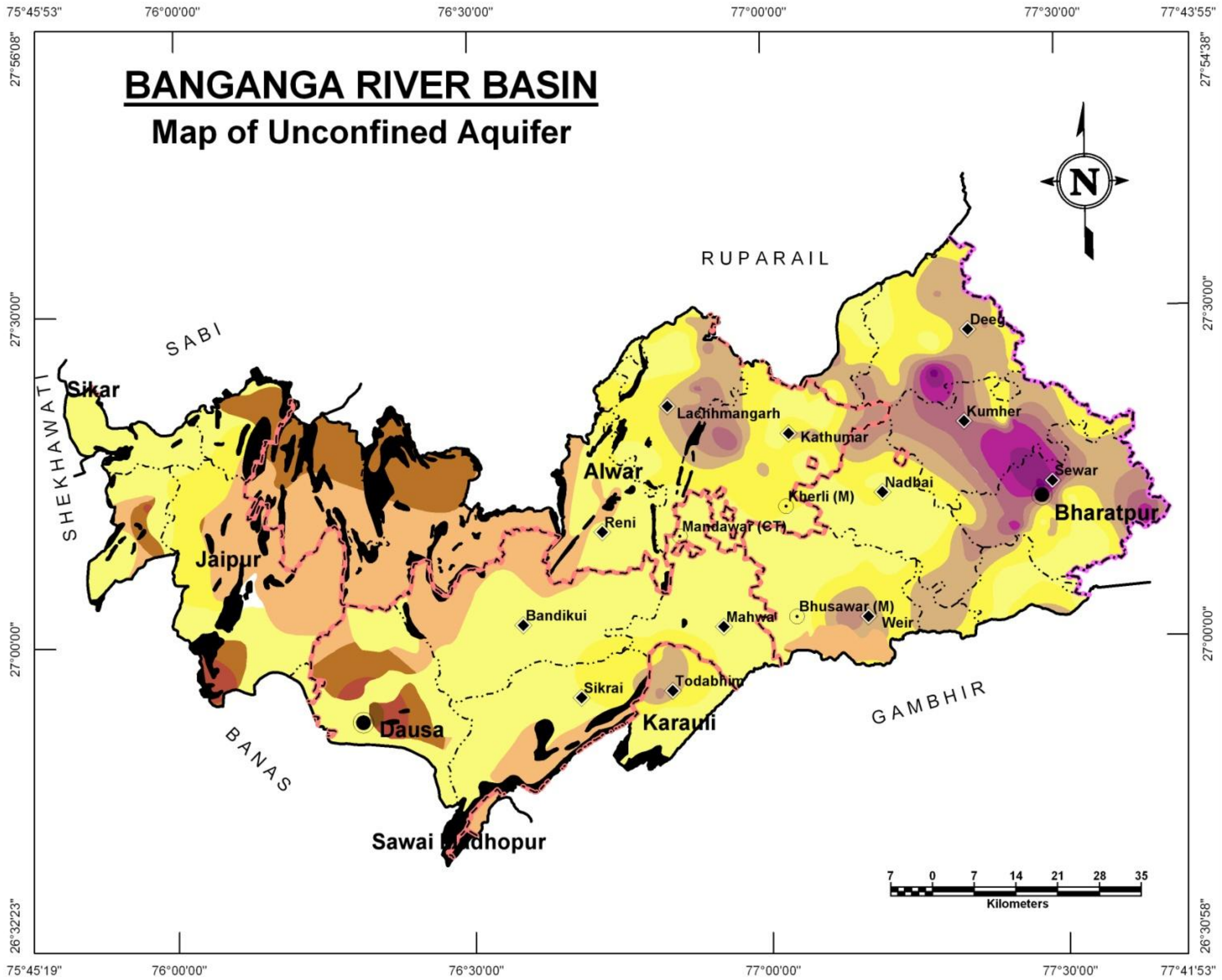
River Basin Boundary

Hills

Depth to Bedrock (m bgl):

- < 40
- 40-60
- 60-80
- 80-100
- > 100

PLATE - XVII



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

River Basin Boundary

Hills

Aquifer Thickness (m):

Alluvial Area:

- < 10
- 10-20
- 20-30
- 30-40
- 40-50
- 50-60
- 60-70
- > 70

Weathered & Fractured Zone in Hardrock Area :

- < 10
- 10-20
- 20-30
- > 30

CONFINED AQUIFERS

First Confined Aquifer

Thick and regionally persistent clay formations separate the unconfined aquifer with the next sandy formation below the unconfined layer of aquifer, forms the first semi-confined to confined aquifer (Plate XVIII) in the region. This aquifer is not found in the western part of the area and largely restricted in occurrence to the eastern part of the hilly region (Bandikui, Reni, Lachhmngarh) attaining greater thicknesses eastwards. The thickness of this aquifer in general, is in the range of <10 to 20 meters in most of the area however with pockets around Sewer, Bharatpur, Nadbai, west of Mahwa and north of Kathumar show a thickness of upto 30 meters. Near Uchain (south of Bharatpur), very high thickness has been noticed.

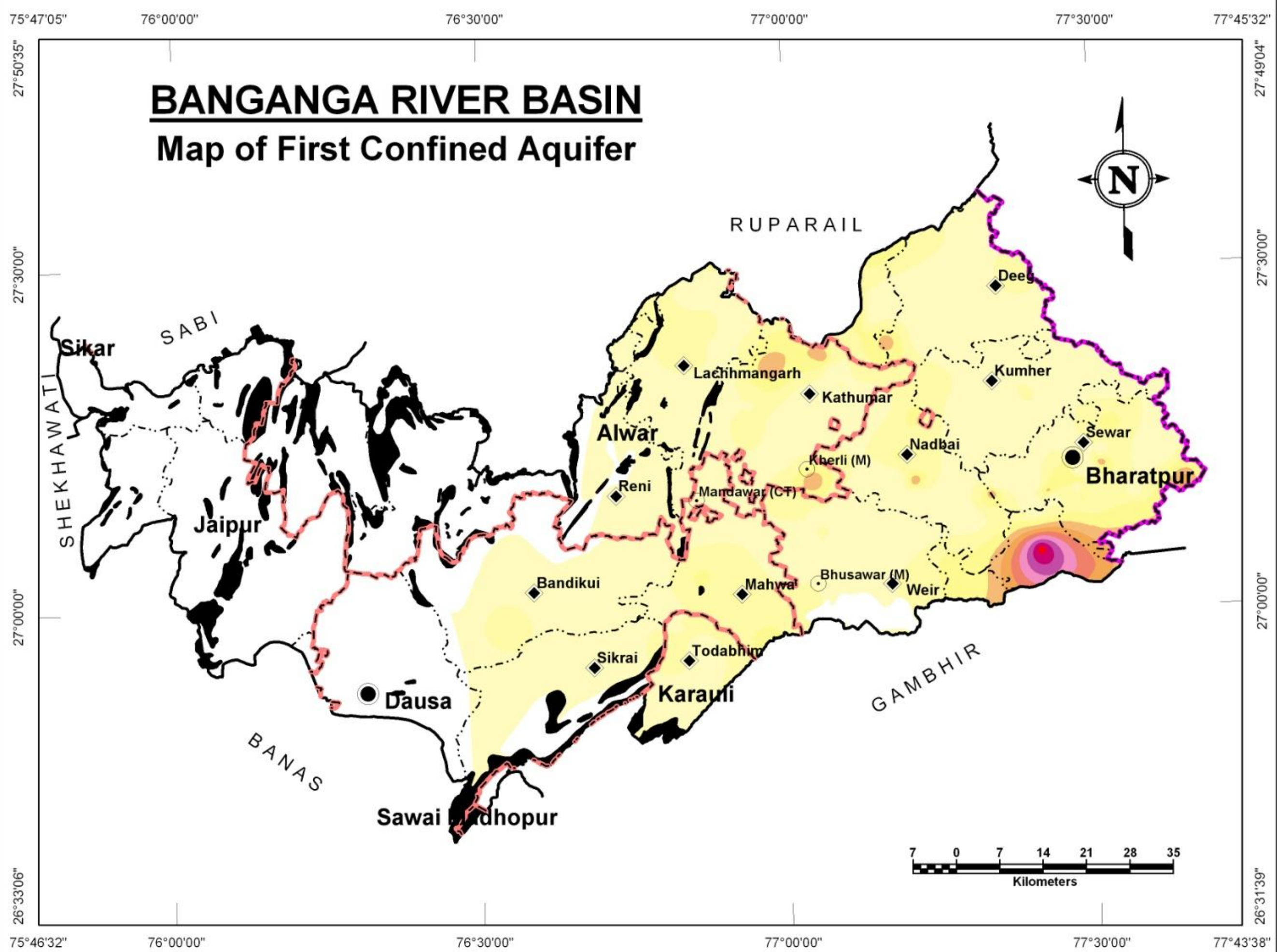
Aquifer Thickness (m)	District Area coverage (sq km)							Total Area (sq km)
	Alwar	Bharatpur	Dausa	Jaipur	Karauli	Sawai Madhopur	Sikar	
< 10	858.72	1,515.57	932.00	-	131.89	-	-	**2,316.8**
10-20	293.42	701.44	268.64	-	75.45	-	-	**287.9**
20-30	87.03	244.84	20.56	-	0.66	-	-	**190.2**
30-40	31.07	72.32	-	-	-	-	-	**145.1**
40-50	0.20	49.40	-	-	-	-	-	**103.0**
50-60	-	34.74	-	-	-	-	-	**95.0**
60-70	-	25.67	-	-	-	-	-	**69.6**
70-80	-	19.04	-	-	-	-	-	**42.2**
80-90	-	8.94	-	-	-	-	-	**25.5**
90-100	-	1.57	-	-	-	-	-	**3,275.3**
Total	**421.34**	**2,667.91**	**185.89**	**-**	**0.12**	**-**	**-**	**2,316.8**

Second Confined Aquifer

The third aquifer of the region i.e., second confined aquifer, occurs further below the 1st confined aquifer, separated by a clayey horizon. This aquifer is also very limited in spatial distribution and occurs only in the far eastern part of the region and largely absent in the central and western part. The general thickness of this is in the range of 10 meters, however in the region of Weir, Uchain, Bharatpur and further northeast, it attains higher thicknesses reaching more than 45m.

Aquifer Thickness (m)	District Area coverage (sq km)							Total Area (sq km)
	Alwar	Bharatpur	Dausa	Jaipur	Karauli	Sawai Madhopur	Sikar	
< 5	404.51	1,749.19	162.93	-	0.12	-	-	**2,316.8**
5-10	11.30	258.96	17.59	-	-	-	-	**287.9**
10-15	3.71	181.14	5.37	-	-	-	-	**190.2**
15-20	1.81	143.29	-	-	-	-	-	**145.1**
20-25	-	103.01	-	-	-	-	-	**103.0**
25-30	-	95.04	-	-	-	-	-	**95.0**
30-35	-	69.56	-	-	-	-	-	**69.6**
35-40	-	42.21	-	-	-	-	-	**42.2**
> 45	-	25.50	-	-	-	-	-	**25.5**
Total	**421.34**	**2,667.91**	**185.89**	**-**	**0.12**	**-**	**-**	**3,275.3**

PLATE -XVIII



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

River Basin Boundary

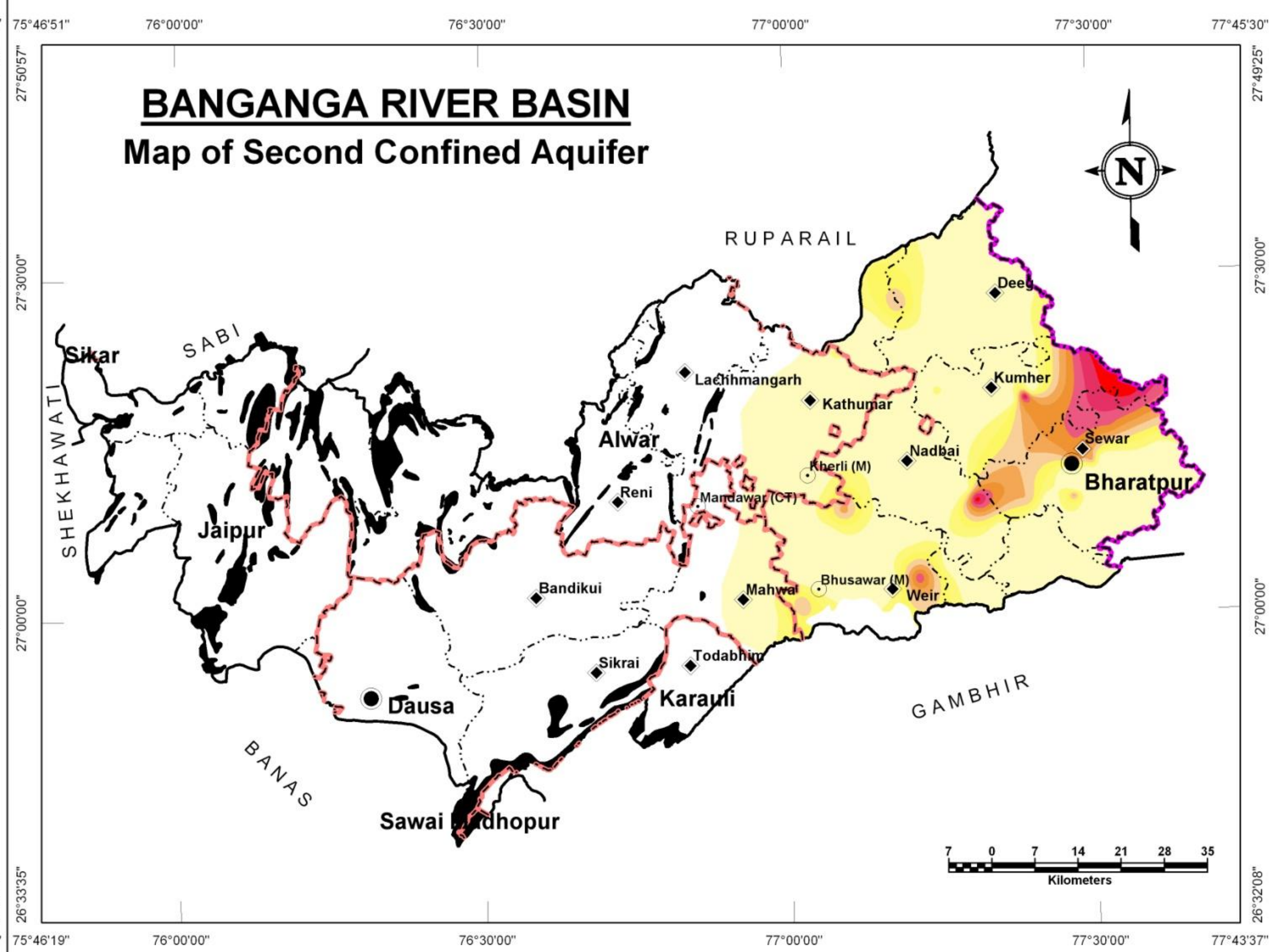
Hills

Aquifer Thickness(m):

Alluvial Areas

- < 10
- 10-20
- 20-30
- 30-40
- 40-50
- 50-60
- 60-70
- 70-80
- 80-90
- 90-100

PLATE - XIX



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

River Basin Boundary

Hills

Aquifer Thickness(m):

Alluvial Areas

- < 5
- 5-10
- 10-15
- 15-20
- 20-25
- 25-30
- 30-35
- 35-40
- 40-45

Four different hydrogeologic cross sections have been prepared to better understand sub-surface lighologic distribution. This has been overlaid with water table elevation of pre-monsoon 2010 and structural faults. The alignment of the cross sections is shown in Plate XX and corresponding sections are presented in Plates XXI through XXIV. The broad orientation of the sections is as given below:

Name of Section Line	Orientation
Section AA’	SW – NE
Section BB’	W – E
Section CC’	S – N
Section DD’	SE – NW

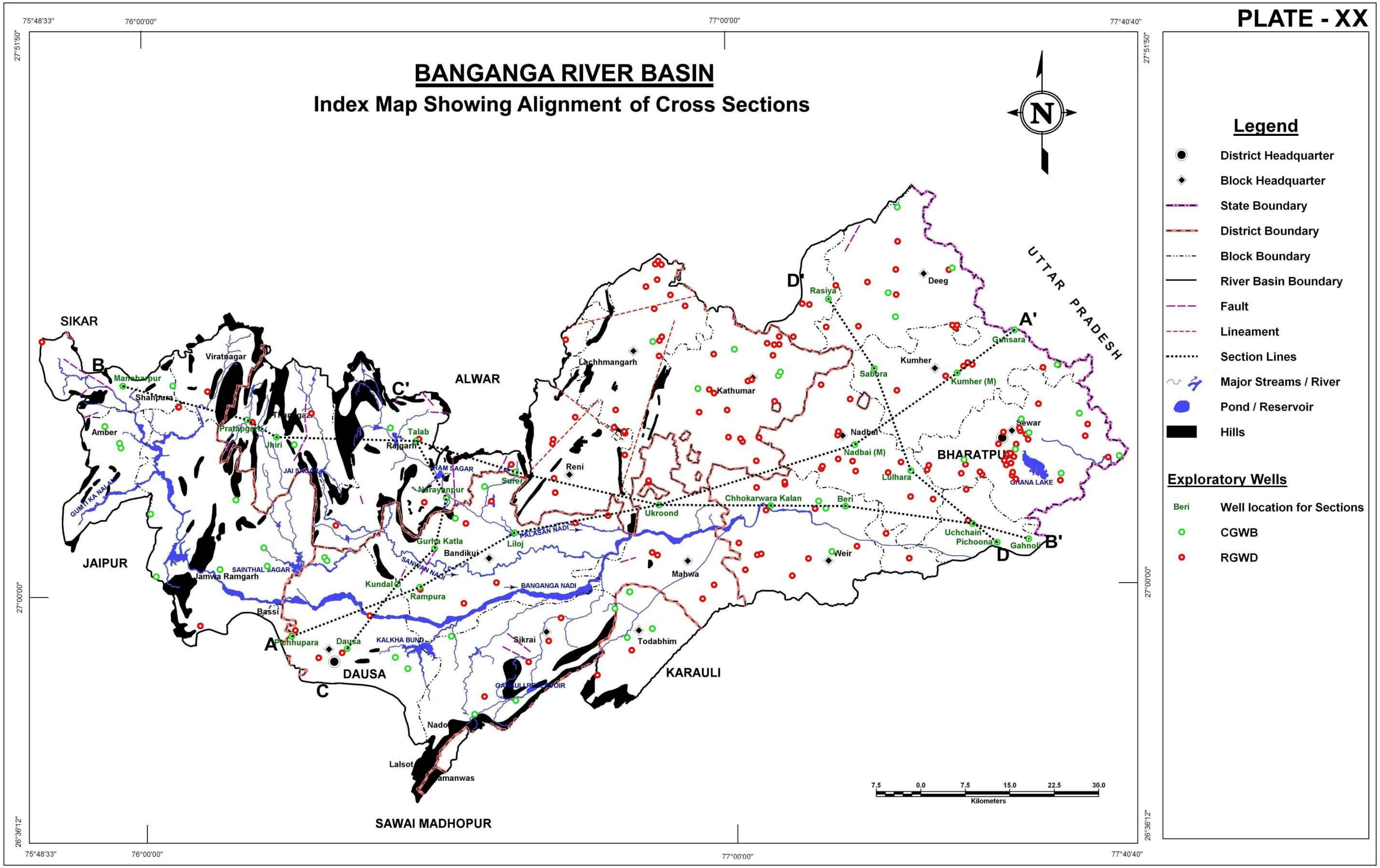


PLATE - XX
BANGANGA RIVER BASIN
Index Map Showing Alignment of Cross Sections
Legend
District Headquarter
Block Headquarter
State Boundary
District Boundary
Block Boundary
River Basin Boundary
Fault
Lineament
Section Lines
Major Streams / River
Pond / Reservoir
Hills
Exploratory Wells
Beri
Well location for Sections
CGWB
RGWD
SIKAR
JAIPUR
ALWAR
DAUSA
KARAULI
SAWAI MADHOPUR
UTTAR PRADESH
BHARATPUR
A
A'
B
B'
C
C'
D
D'
Viratnagar
Shahpura
Amber
Jamwa Ramgarh
Bassi
Lachhmangarh
Kathumar
Deeg
Kumher
Nadbai
Reni
Rajgarh
Mahwa
Todabhim
Sikrai
Lalsot
Weir
Rasiya
Sabora
Kumher (M)
Nadbai (M)
Lulhara
Gunsara
Sewar
Chhokarwara Kalan
Beri
Ukroond
Uchchain
Pichoona
Gahnoli
Talab
Jhiri
Pratapgarh
Surer
Narayanpur
Gurha Katla
Bandikui
Liloj
Kundal
Rampura
Dausa
Pichhupara
Nadoti
Bamanwas
RAM SAGAR
JAI SAGAR
SAINTHAL SAGAR
KALKHA BUND
GHANA LAKE
BANGANGA NADI
PALASAN NADI
SANWAN NADI
GUMTI KA NALA
7.5
0.0
7.5
15.0
22.5
30.0
Kilometers
75°48'33"
76°00'00"
77°00'00"
77°40'40"
27°51'50"
27°00'00"
26°36'12"

Section A-A':

This section is the longest of the sections plotted in the area and trends in SW-NE direction, cutting across the basin (Plate XXI). The section depicts the disposition of different layers of sand and clay along with weathered and fractured zones in limestone, schist, phyllite and quartzite. The south western part of the river basin is predominantly hard rock with aquifer formed within quartzite and schist. Due to availability of lithologs of limited depths the continuity of the same is not observed as moved towards north east. However, few deep lithologs in the NE part have shown the presence of schist and quartzite. In the rest of the area, sand is predominating while there are some lenses of clay. The water level varies from 174 m amsl to 323 m amsl following the surface topography as observed from the 2010 pre monsoon season.

Section B-B':

The B-B' section is another long section trending SW-NE (Plate XXII). The section depicting sand and clay layers of alluvium aquifers underlined by thick zone of weathered and fractured schist, quartzite and granite. In the central part of the section is pre dominated by granite and quartzite. The alternate sequence of clay and sand is forming the major aquifer system in these areas. In some of the wells towards the NE of the cross section quartzites are encountered with limited thickness of weathered and fractured zone. Limited depth of the wells and their lithologs information below it is not available. However, from the cross section it is apparent that occurrence of hard rock aquifer in form of weathered and fractured quartzite is occurring below the alluvium.

Thickness of alluvium in North eastern part more than 150 m and decreases to around 100 m in the central part of the river basin. It is observed from pre monsoon 2010 data the Ground water level varies from 173 m amsl to 460 m amsl from SW – NE of the section.

There is a major fault indication observed in south western part of the basin in between phyllite, limestone and schist.

PLATE - XXI

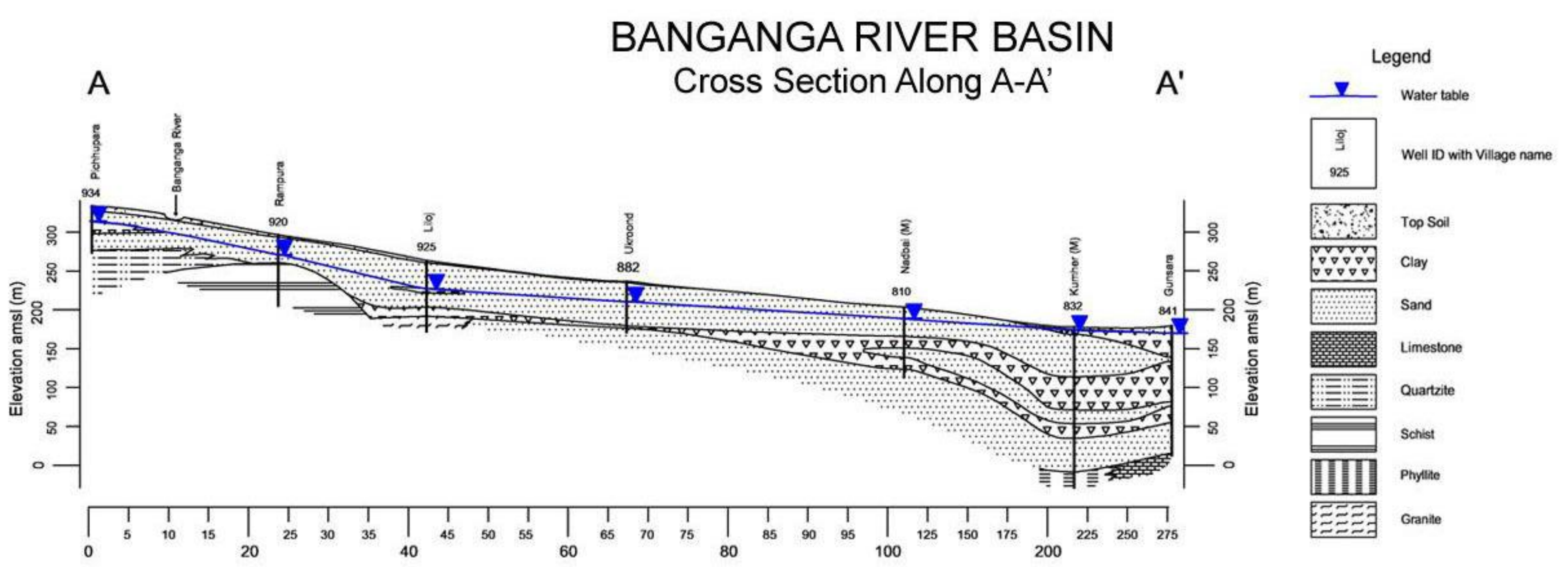
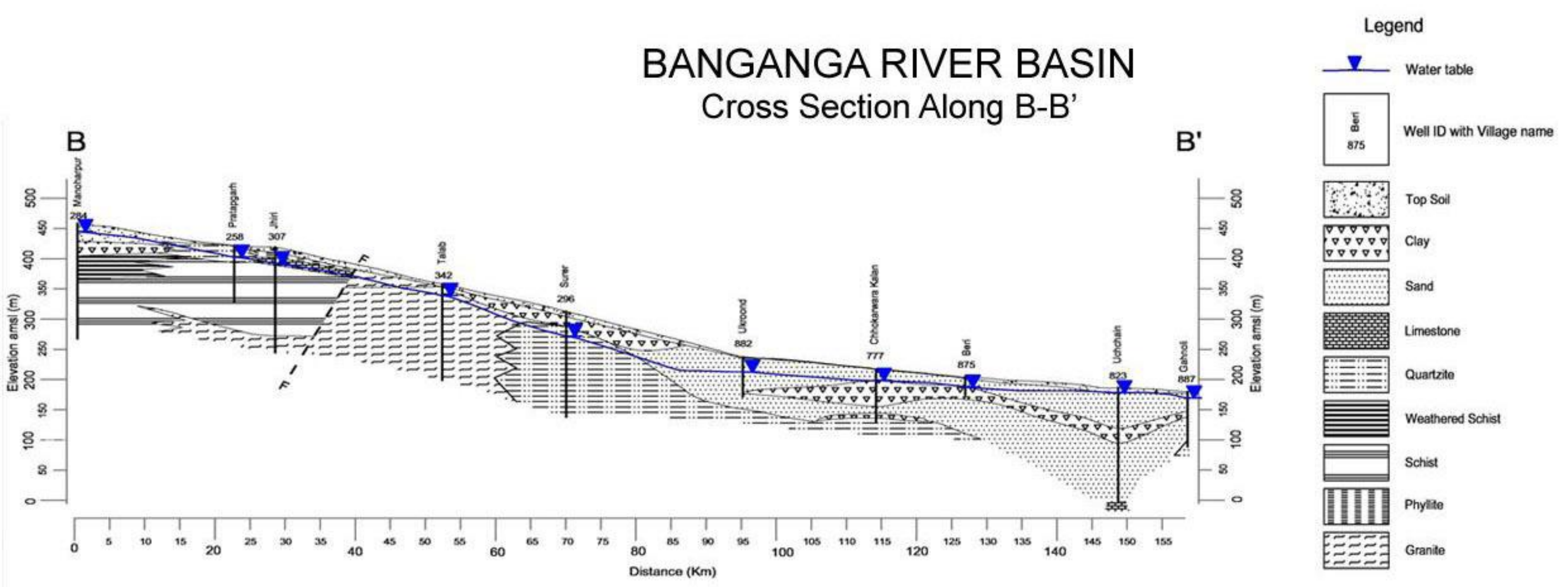


PLATE - XXII



Section C-C':

The C-C' section has been selected across the basin trending SW – NE in the western part of the basin. The northern part of the section is characterized by the hard rock formation of Granite. The weathered and fractured portions of quartzite are most suitable aquifer zones. Cross section (Plate XXIII) depicts, that in southern part, alluvium is forming the major aquifer system underlain by weathered schist, which is saturated with water.

In this section the thickness of alluvium ranges from 50 to 80 m however the hard rock aquifer are explored limited to the thickness of 30 m.

The water level varies from 267 m amsl to 322 m amsl as per the data of pre monsoon 2010.

Section D-D':

This section is selected in the central part of the basin which depicts mainly the alternating layers of clay and sand with hard rock being encountered in some of the wells. As all the wells have not been drilled till basement the limited information available has been interpolated for preparation of the sections. Thickness of alluvium in this area ranges from 100 m to more than 150 m (Plate XXIV). In the eastern part of the area quartzite is encountered in the well.

It is observed from pre monsoon 2010 data the ground water level varies from 177 m amsl to 183 m amsl. In this area alluvium acts as semi confined aquifer due to continuous and thick impermeable layer of clay mixed with kankar overlying on it and encountered in all the wells in the area.

A fault zone appears to pass through this section as indicated.

PLATE - XXIII

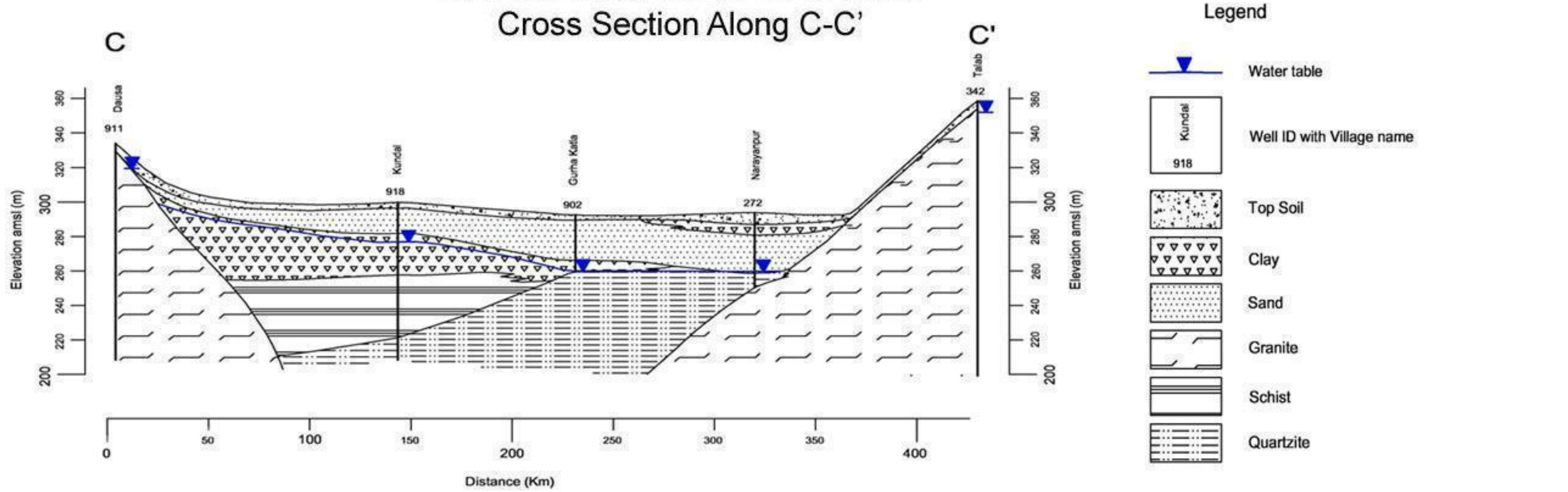
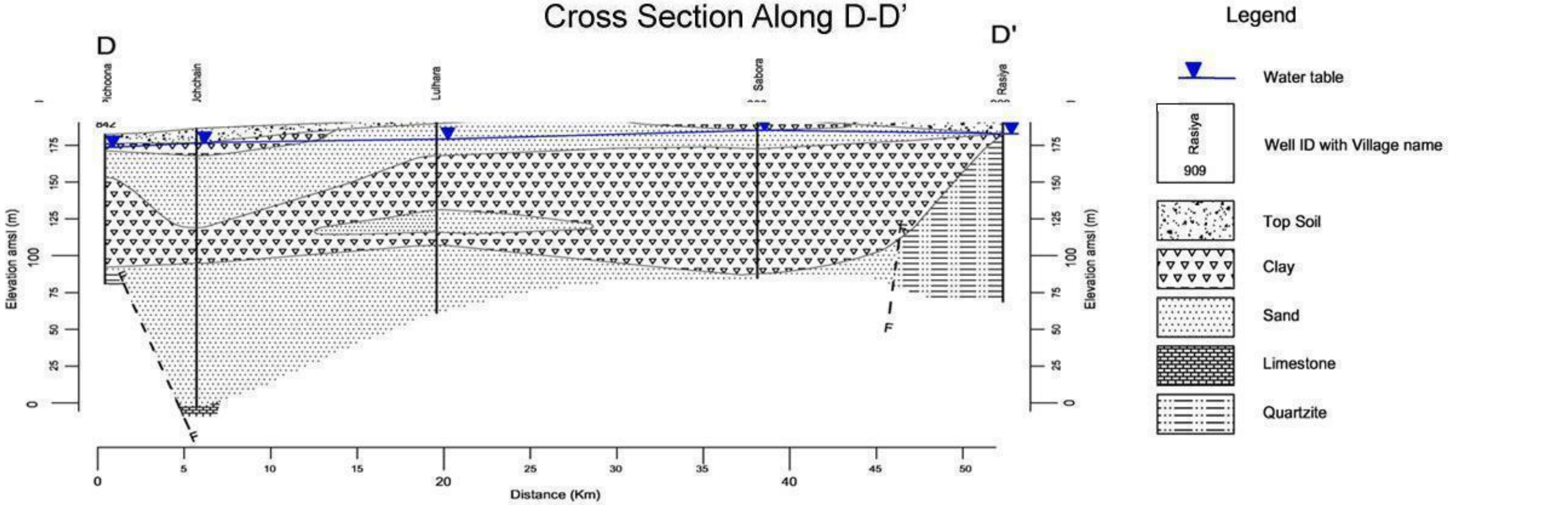


PLATE - XXIV



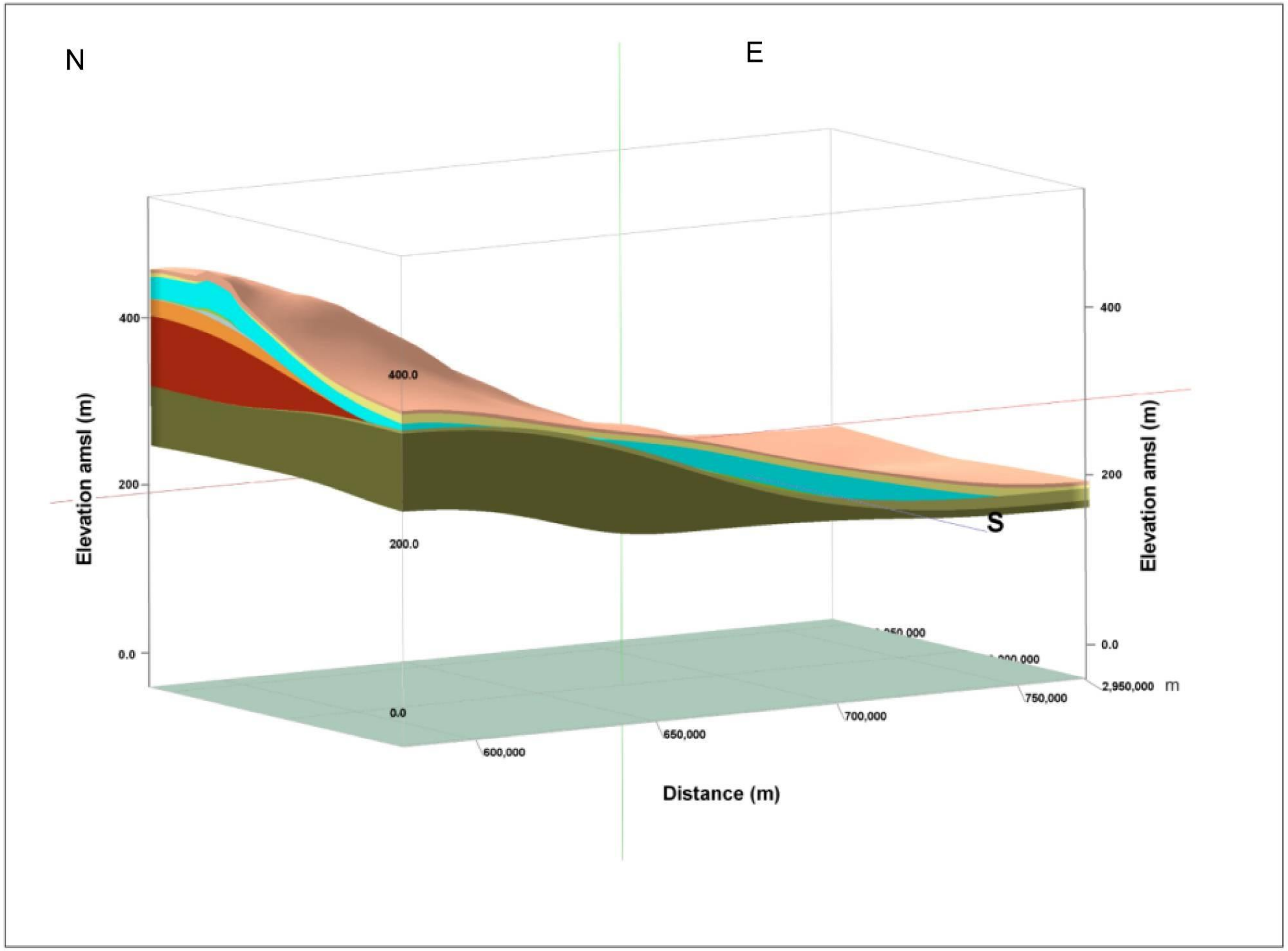
A 3D model depicting the various litho-stratigraphic units in the entire river basin is presented in Plate XXV. The model reveals that beneath the first sandy unconfined aquifer, there are two persistent clay horizons (1st largely persistent all over the alluvium to the east of hilly area whereas the 2nd one largely confined to far eastern part of the basin only) in the region separating three sandy aquifers. The second sandy aquifer is found only in the eastern part of the basin within alluvium whereas the third sandy aquifer is present only to a limited extent in the eastern part of alluvial terrain.

The depth of hard rock is shallow in the western part as compared to the eastern part of the basin. Granite Gneiss is the basement rock occurring in the western part of the basin which is not been encountered in eastern part. The small discontinuous patches depicted in 3D model below the second sandy aquifer are indicative of occurrence of limestone, quartzite, phyllite, slate and schist etc. as reported in data of GWD and CGWB. In absence of continuity of the similar formation in adjacent well data they are occurring as small patches. However, the quartzite, below the second sandy layer is appearing as almost continuous layer in the model.

BANGANGA RIVER BASIN

3D Model

PLATE - XXV



LEGEND

- Top Soil
- Clay
- Unconfined aquifer in alluvium
- Clay
- First confined aquife in alluvium
- Clay
- Second confined aquifer in alluvium
- Schist
- Weathered & Fractured Slate
- Slate
- Weathered & Fractured Phyllites
- Phyllites
- Weathered & Fractured Limestone
- Limestone
- Weathered & Fractured Schist
- Schist
- Weathered & Fractured Quartzite
- Quartzite
- Weathered & Fractured Granite/Gniess
- Granite/Gniess

Glossary of terms

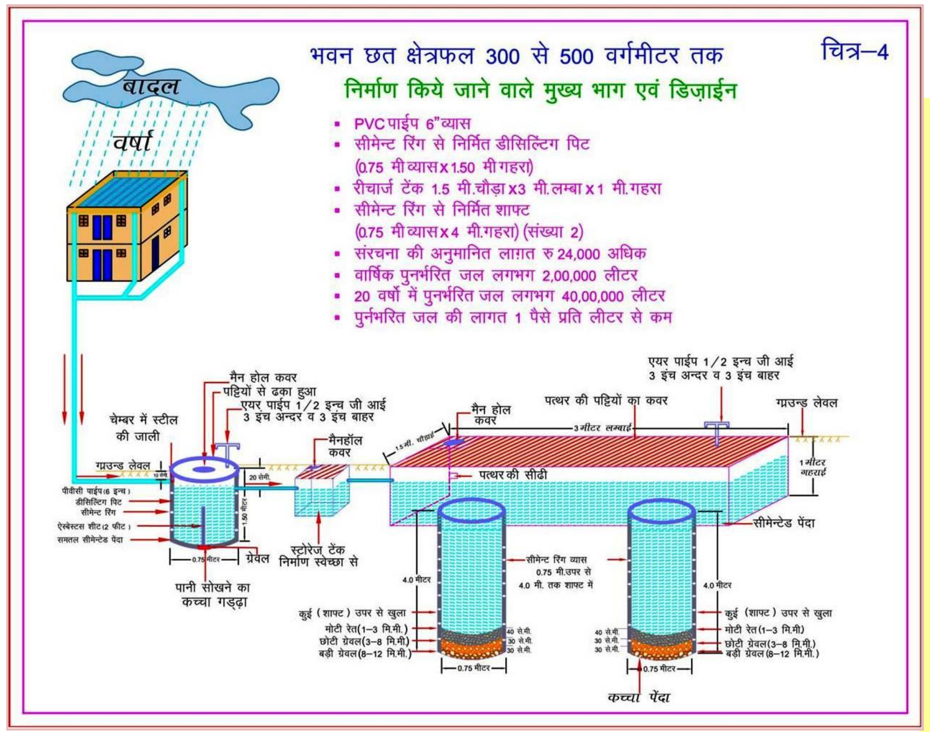
S. No.	Technical Terms	Definition
1	AQUIFER	A saturated geological formation which has good permeability to supply sufficient quantity of water to a Tube well, well or spring.
2	ARID CLIMATE	Climate characterized by high evaporation and low precipitation.
3	ARTIFICIAL RECHARGE	Addition of water to a groundwater reservoir by man-made activity
4	CLIMATE	The sum total of all atmospheric or meteorological influences principally temperature, moisture, wind, pressure and evaporation of a region.
5	CONFINED AQUIFER	A water bearing strata having confined impermeable overburden. In this aquifer, water level represents the piezometric head.
6	CONTAMINATION	Introduction of undesirable substance, normally not found in water, which renders the water unfit for its intended use.
7	DRAWDOWN	The drawdown is the depth by which water level is lowered.
8	FRESH WATER	Water suitable for drinking purpose.
9	GROUND WATER	Water found below the land surface.
10	GROUND WATER BASIN	A hydro-geologic unit containing one large aquifer or several connected and interrelated aquifers.
11	GROUNDWATER RECHARGE	The natural infiltration of surface water into the ground.
12	HARD WATER	The water which does not produce sufficient foam with soap.
13	HYDRAULIC CONDUCTIVITY	A constant that serves as a measure of permeability of porous medium.
14	HYDROGEOLOGY	The science related with the ground water.
15	HUMID CLIMATE	The area having high moisture content.
16	ISOHYET	A line of equal amount of rainfall.
17	METEOROLOGY	Science of the atmosphere.
18	PERCOLATION	It is flow through a porous substance.
19	PERMEABILITY	The property or capacity of a soil or rock for transmitting water.
20	pH	Value of hydrogen-ion concentration in water. Used as an indicator of acidity ($pH < 7$) or alkalinity ($pH > 7$).
21	PIEZOMETRIC HEAD	Elevation to which water will rise in a piezometers.
22	RECHARGE	It is a natural or artificial process by which water is added from outside to the aquifer.
23	SAFE YIELD	Amount of water which can be extracted from groundwater without producing undesirable effect.
24	SALINITY	Concentration of dissolved salts.
25	SEMI-ARID	An area is considered semiarid having annual rainfall between 10-20 inches.
26	SEMI-CONFINED AQUIFER	Aquifer overlain and/or underlain by a relatively thin semi-pervious layer.
27	SPECIFIC YIELD	Quantity of water which is released by a formation after its complete saturation.
28	TOTAL DISSOLVED SOLIDS	Total weight of dissolved mineral constituents in water per unit volume (or weight) of water in the sample.

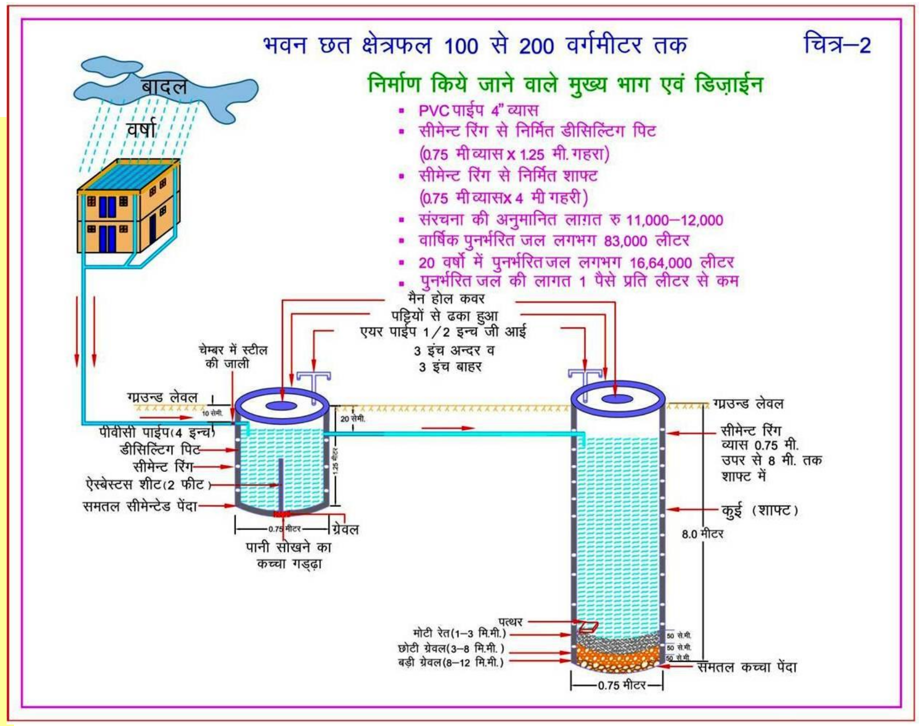
(Contd...)

S. No.	Technical Terms	Definition
29	TRANSMISSIBILITY	It is defined as the rate of flow through an aquifer of unit width and total saturation depth under unit hydraulic gradient. It is equal to product of full saturation depth of aquifer and its coefficient of permeability.
30	UNCONFINED AQUIFER	A water bearing formation having permeable overburden. The water table forms the upper boundary of the aquifer.
31	UNSATURATED ZONE	The zone below the land surface in which pore space contains both water and air.
32	WATER CONSERVATION	Optimal use and proper storage of water.
33	WATER RESOURCES	Availability of surface and ground water.
34	WATER RESOURCES MANAGEMENT	Planned development, distribution and use of water resources.
35	WATER TABLE	Water table is the upper surface of the zone of saturation at atmospheric pressure.
36	ZONE OF SATURATION	The ground in which all pores are completely filled with water.
37	ELECTRICAL CONDUCTIVITY	Flow of free ions in the water at 25C mu/cm.
38	CROSS SECTION	A Vertical Projection showing sub-surface formations encountered in a specific plane.
39	3-D PICTURE	A structure showing all three dimensions i.e. length, width and depth.
40	GWD	Ground Water Department
41	CGWB	Central Ground Water Board
42	CGWA	Central Ground Water Authority
43	SWRPD	State Water Resources Planning Department
44	EU-SPP	European Union State Partnership Programme
45	TOPOGRAPHY	Details of drainage lines and physical features of land surface on a map.
46	GEOLOGY	The science related with the Earth.
47	GEOMORPHOLOGY	The description and interpretation of land forms.
48	PRE MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer before Monsoon (carried out between 15th May to 15th June)
49	POST-MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer after Monsoon (carried out between 15th October to 15th November)
50	PIEZOMETER	A non-pumping small diameter bore hole used for monitoring of static water level.
51	GROUND WATER FLUCTUATION	Change in static water level below ground level.
52	WATER TABLE	The static water level found in unconfined aquifer.
53	DEPTH OF BED ROCK	Hard & compact rock encountered below land Surface.
54	G.W. MONITORING STATION	Dug wells selected on grid basis for monitoring of state water level.
55	EOLIAN DEPOSITS	Wind-blown sand deposits

Ground Water Department, Rajasthan

European Union
State Partnership Programme



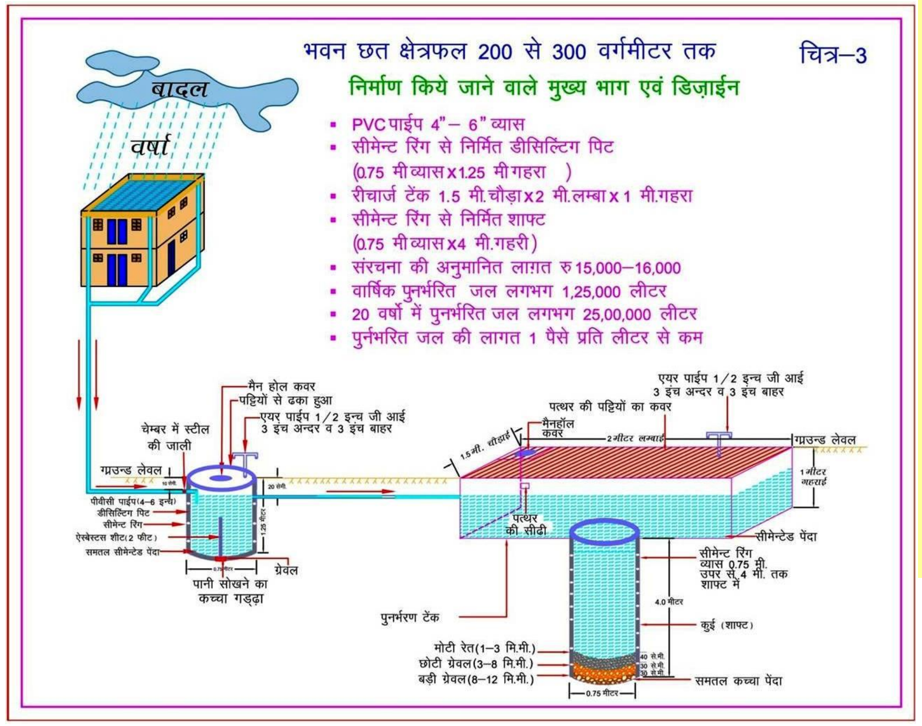


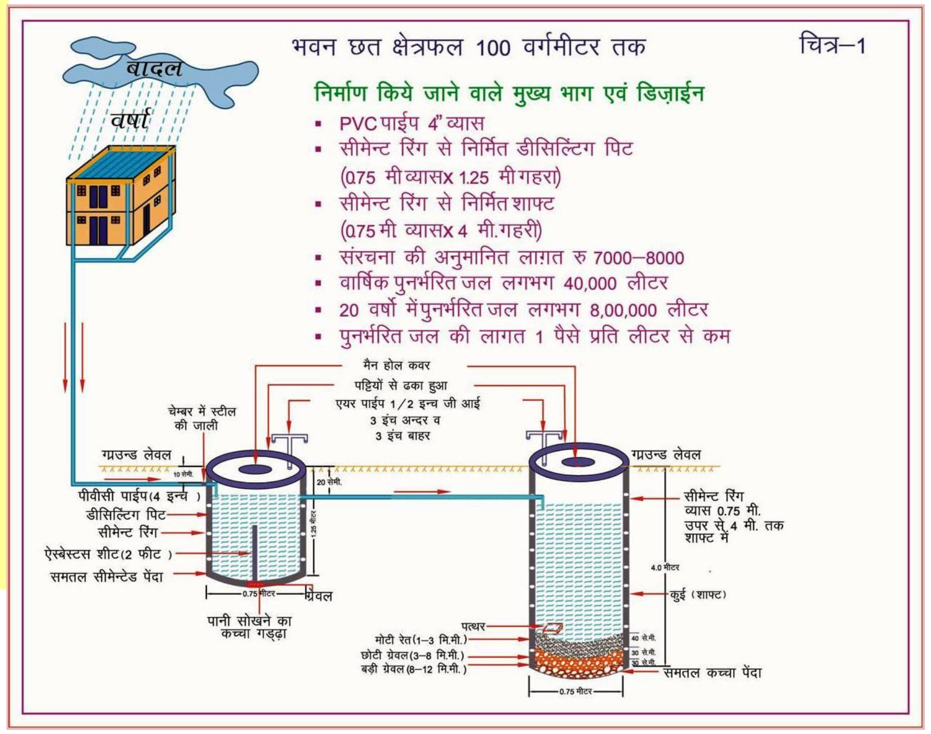
भूजल में घुले मुख्य तत्वों की अधिकता का मानव शरीर पर दुष्प्रभाव

- बोरोन-स्नायु तन्त्र पर प्रभाव
- क्लोराइड-सोडियम के साथ मिलकर उच्च रक्त चाप
- कैल्शियम-जोड़ों में कड़ापन
- आर्सैनिक-त्वचा रोग, कैंसर
- लेड-बच्चों के शारीरिक व मानसिक विकास में बाधा वयस्कों में गुर्दे के रोग
- फ्लोराइड - दंत क्षरण
- सोडियम-हृदय, गुर्दा व रक्त परिसंचरण रोगों से ग्रसित लोगों को हानिकारक
- नाइट्रेट-नवजात शिशुओं में ब्लू बेबी बीमारी (मेथमोग्लोबिनिमिया)
- सल्फेट-अधिकता में मैग्नेशियम के साथ मिलकर दस्तावर
- आयरन-आयरन जीवाणु से आमाशय संबंधी रोग
- फ्लोराइड-जोड़ों में अकड़न, हड्डियों में मुड़ाव

केन्द्रीय भूमि जल बोर्ड,
पश्चिमी क्षेत्र, जयपुर
जल संसाधन मंत्रालय
भारत सरकार
e-mail:cgwbwr@sancharnet.in

भूजल अमूल्य है इसे प्रदूषित न करें।





Rolta India Limited

Myths and Facts about Ground Water

S No	Myths	Facts
1	What is Ground Water • an underground lake • a net work of underground rivers • a bowl filled with water	Water which occurs below the land in geological formations/rocks is Ground water
2	Ground Water occurs everywhere beneath the Land Surface	Not really, it depends on the nature of rock formation
3	There is a relationship between ground water and surface water	Not all the places. Near streams/rivers there is relation
4	Groundwater is not renewable resource	It is renewable source and every year it is being recharged through rain/applied irrigation etc
5	Ground water is unlimited and deeper you drill more discharge	It is limited to annual recharge from rain/applied irrigation. The discharge may not increase if you go deeper
6	Ground Water moves rapidly	The movement of ground water is very slow
7	Ground water pumped from wells is thousands of years old	Generally the ground water being tapped through wells is a few years old
8	If water taste good—it is safe to drink	It may have other chemicals e.g. fluoride, nitrates etc which are harmful
9	Water from free flowing tube wells is very pure	This water can also be contaminated so test before use
10	If I recharge my TW/DW/HP it will not benefit me	It will also benefit you and also adjoing wells
11	There is no static ground water resources in Rajasthan	Rajasthan is also having Static GW resources, and being tapped in most of areas as GW annual withdrawal is more than annual recharge
12	I cannot meet annual cooking and drinking water requirement by rain water harvesting	The water requirement for drinking and cooking is only 8 lit/day. You can harvest this water for family of 5 persons from roof top or paved area of 75 Sq m to meet annual requirement
13	You can increase ground water recharge	This can be done by harvesting the rain water and storing in sub surface reservoir (GW) by constructing the recharge structures
14	You cannot use abandoned TW/HP/DW for ground water recharge	These should be used as recharge structures as harvested rain water is directly put into GW reservoir
15	Putting waste near HP/TW will not cause any problem	Such actions will pollute wells and water

ROLTA

Rolta India Limited

Central & Registered Office
Rolta Tower A,
Rolta Technology Park,
MIDC, Andheri (East),
Mumbai - 400 093
Tel : +91 (22) 2926 6666, 3087 6543
Fax : +91 (22) 2836 5992
Email : indsales@rolta.com

www.rolta.com